Stone–Jordan Exceptional Theory

From the Void by Mirror Climb

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Abstract

We formulate *Stone–Jordan Exceptional Theory* (SJET) from two primitive axioms: the **void** $\mathcal{V}$ (no observable distinction) and a **mirror operator** M that generates structure by involutive doubling. Iterating M on the Stone spectrum of $\mathcal{V}$ climbs the Hurwitz ladder $0 \to \mathbb{R} \to \mathbb{C} \to \mathbb{H} \to \mathbb{O} \to J_3(\mathbb{O})$ and the exceptional cascade $0 \to \mathrm{E}_8 \to \mathrm{E}_8 \times \mathrm{E}_8 \to \mathrm{E}_6 \to \cdots \to 0$; capacity balancing on the mirror-refined graph is the *derived* selection principle, not an independent axiom. At the E_6 rung we prove vacuum selection on EIII, composite $\mathrm{Spin}(10) \times U(1)$ gauge theory with anomaly cancellation, and induced gravitation; postulates P1, P2, P4, P5 complete SM descent (three families derived; P1/P2 not promotable from mirror). Conditional ToE closure is recorded in Section 10.1. The universe is the stabilized mirror climb $\mathsf{C}^\infty(\mathcal{V})$, equivalently the capacity fixed point on the Albert–Cayley graph.

Keywords: void; mirror operator; Stone duality; exceptional Jordan algebra; E_6 coset; induced gravity.
MSC 2020: 03G05, 17C27, 22E46, 81T13.

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1 Introduction

SJET asks: what is the minimal mathematical seed from which the Standard Model and gravitation can be *derived*, not merely embedded? Our answer is two-fold:

Axiom 1 The void $\mathcal{V}$ — no observable distinction, the terminal object of the Stone ladder.

Axiom 2 The mirror operator M — an involutive generator that, applied repeatedly, *climbs* the division and exceptional ladders.

Everything else — graphs, Albert algebra, capacity dual, E_6 potential, gauge sector, induced gravitation, anomaly cancellation — is **derived** from $\mathsf{M}^n(\mathcal{V})$ and capacity balancing on mirror-refined partitions. Physical postulates P1, P2, P4, P5 (soldering, chiral matter, GUT–Higgs, reflection positivity) remain the honest boundary between proved mathematics and four-dimensional quantum field theory; three families are **derived** (Theorem 8.6), not postulated. P1 and P2 were checked for derivation from mirror stabilisation and *not* promoted (Propositions 6.14, 8.9). Section 10.1 records the conditional ToE closure ledger.

1.1 The climb picture

Schematically:

$$\mathcal{V} \xrightarrow{\mathsf{M}} S_{\mathbb{R}} \xrightarrow{\mathsf{M}} S_{\mathbb{C}} \xrightarrow{\mathsf{M}} S_{\mathbb{H}} \xrightarrow{\mathsf{M}} S_{\mathbb{O}} \xrightarrow{\mathsf{M}} J_3(\mathbb{O}) \xrightarrow{\mathsf{M}} \cdots \xrightarrow{\mathsf{M}} \mathcal{V}. \tag{1.1}$$

The bottom and top 0 are the same void: the climb is a *closed* ladder. At each rung, M acts as Boolean complement (discrete), Cayley–Dickson doubling (normed division), or heterotic/Cartan doubling (exceptional Lie). Capacity balancing $\mathcal{C}$ is the *dual mirror*: it selects the unique balanced partition at each level (BAMLP), not a separate axiom.

1.2 Epistemic conventions

- **Theorem** — proved from Axioms 1–2 and standard mathematics.
- **Derived** — explicit construction from mirror climb (no extra postulate).
- **Postulate** — added four-dimensional physics (P1, P2, P4, P5, CC), named once.
- **Conjecture** — motivated but not controlled.

1.3 Conventions

Signature $(-,+,+,+)$; compact E_6 for dynamics; $\Phi \in J_3(\mathbb{O})$ order parameter; $\mathcal{C}$ capacity functional; M mirror operator; $\mathcal{V}$ void. Casimirs: $C_A(\mathrm{Spin}(10)) = 8$, $T(\mathbf{16}) = 2$, $C_2(\mathbf{16}) = 45/8$. Branchings as in (3.12).

1.4 Organization

Section 2: void and mirror axioms; functor M; Theorem 2.8 ($\mathsf{M}^4(\mathcal{V}) = \mathbb{O}$, $\mathsf{M}^5(\mathcal{V}) = J_3(\mathbb{O})$); Theorems 3.10–3.12 ($\mathsf{M}^6(\mathcal{V}) = E_8$, $\mathsf{M}^7(\mathcal{V}) = E_8 \times E_8$). Section 3: exceptional continuation and capacity dual. Section 4: derived graph, Stone lift, and order parameter. Section 5: capacity, potential, vacuum. Sections 6–8: physics under P1–P4. Section 9: ledger. Appendix A: Albert identities.

2 The Void and the Mirror Operator

This section states the *only* primitive axioms of SJET and gives the functorial definition of M on which all rung theorems depend.

2.1 The void

Definition 2.1 (Void)**.** The *void* $\mathcal{V}$ is the trivial observable algebra: the two-element Boolean algebra $\mathcal{B}_0 = \{\bot, \top\}$ with $\bot < \top$, and its Stone spectrum $S_0 = \mathrm{Spec}(\mathcal{B}_0)$, a single point. The associated division datum is the zero algebra $\{0\}$ of dimension 0. No vertex, no edge, no localisable quantum number exists at $\mathcal{V}$.

Axiom 2.2 (Void)**.** Physical reality is anchored at the void $\mathcal{V}$: the state with no distinguishable observable. All structure arises by departure from $\mathcal{V}$, not by importing a graph or Lie group as primitive data.

2.2 Category of Stone rungs

Definition 2.3 (Stone rung category)**.** Let **Stone** be the category of Stone spaces (compact, Hausdorff, totally disconnected) with continuous maps. A *rung datum* is a pair (S, D) with $S \in \mathbf{Stone}$ and D a formally real observable algebra at that rung. Morphisms preserve both S and the algebraic rung label.

Definition 2.4 (Three mirror modes)**.** The mirror operator M acts in three modes, triggered by rung index:

M1. Discrete ($n = 0 \to 1$): append a Boolean generator a_{n+1} to $\mathcal{B}_n$, doubling clopen sectors: $\mathcal{B}_{n+1} = \mathcal{B}_n[a_{n+1}]/(a_{n+1}^2 = a_{n+1})$.

M2. Arithmetic ($1 \to 2 \to 3 \to 4$): Cayley–Dickson doubling $D_{n+1} = D_n \oplus D_n$ with product $(x, y) \cdot (u, v) = (xu - \bar{v}y,\, vx + y\bar{u})$ and involution $\overline{(x, y)} = (\bar{x}, -y)$ (Appendix A.1).

M3. Jordan exit ($4 \to 5$): because $\mathsf{M}_{\mathrm{arith}}(\mathbb{O})$ is non-division (sedenions), the climb exits to $J_3(\mathbb{O}) = H_3(\mathbb{O})$, the unique formally real Jordan envelope of $\mathbb{O}$.

M4. Exceptional ($n \geq 6$): Lie mirror $G \mapsto G \times G$ or Cartan involution on G/H (Section 3.3).

Definition 2.5 (Mirror operator)**.** A *mirror operator* is a endofunctor $\mathsf{M} : \mathbf{Stone} \to \mathbf{Stone}$ together with algebra maps such that:

(i) M is **involutive** on stabilized rungs: $\mathsf{M}^2 \cong \mathrm{id}$ (Boolean complement, field conjugation, coset exchange $\mathbf{16} \leftrightarrow \overline{\mathbf{16}}$).

(ii) M is **nontrivial** at $\mathcal{V}$: $S_1 = \mathsf{M}(S_0)$ has $|S_1| = 2$ clopen points.

(iii) M is **functorial**: compatible maps $f : S \to T$ induce $\mathsf{M}(f) : \mathsf{M}(S) \to \mathsf{M}(T)$.

Axiom 2.6 (Mirror)**.** There exists M as in Definition 2.5 whose orbit from $\mathcal{V}$ generates the unified ladder (3.1). At each rung, capacity balancing $\mathrm{Bal}_{\mathcal{C}}$ is the *dual mirror* (Theorem 3.29). The climb is closed: $\mathsf{M}^{\mathrm{tot}}(\cdots(\mathrm{E}_6)\cdots) = \mathcal{V}$.

2.3 Rung table and the Hurwitz–Jordan theorems

Definition 2.7 (Mirror power)**.** Define rung algebras A_n and dimensions d_n by

$$(S_n, A_n) \;=\; \mathsf{M}^n(S_0, A_0), \qquad (S_0, A_0) = (S(\mathcal{V}), \{0\}), \qquad d_n = \dim_{\mathbb{R}} A_n. \tag{2.1}$$

Theorem 2.8 (Mirror rung table)**.** *The first six arithmetic rungs of* $\mathsf{M}^n(\mathcal{V})$ *are:*

n	A_n	d_n	*Stone points* $\lvert S_n\rvert$	*Mirror mode*
0	$\{0\}$	0	1	*void*
1	$\mathbb{R}$	1	2	*discrete*
2	$\mathbb{C}$	2	4	*arithmetic*
3	$\mathbb{H}$	4	8	*arithmetic*
4	$\mathbb{O}$	8	16	*arithmetic (Hurwitz terminal)*
5	$J_3(\mathbb{O})$	27	27	*Jordan exit*

In particular $\mathsf{M}^4(\mathcal{V})$ *carries the octonions and* $\mathsf{M}^5(\mathcal{V}) \cong J_3(\mathbb{O})$*.*

Proof. Discrete step ($n = 0 \to 1$): adjoining one Boolean atom to $\mathcal{B}_0$ gives the four-element algebra $\{\bot, a, \neg a, \top\}$ with Stone spectrum two points; this is the first observable distinction, identified with ordered data $\mathbb{R}$ ($d_1 = 1$).

Arithmetic steps ($n = 1, 2, 3$): Cayley–Dickson doubling doubles dimension: $1 \to 2 \to 4 \to 8$. Hurwitz's theorem (Theorem 3.3) proves no further *normed division* algebra exists at $d = 16$.

Jordan exit ($n = 4 \to 5$): $\mathsf{M}_{\mathrm{arith}}(\mathbb{O})$ is the sedenion algebra $\mathbb{S}$ ($d = 16$), which fails division (Lemma 2.9). The climb therefore exits to the unique formally real Jordan algebra $H_3(\mathbb{O})$ with $d_5 = 27$ and cubic norm $N(X) = \det X$ (Proposition 3.4). The Stone spectrum S_5 has 27 clopen cells matching the 27 matrix coordinates. □

Lemma 2.9 (Sedenion failure)**.** *Cayley–Dickson doubling of* $\mathbb{O}$ *produces* $\mathbb{S}$ *of dimension* 16 *with zero divisors;* $\mathbb{S}$ *is not a division algebra. Hence* $\mathsf{M}^5_{\mathrm{arith}}(\mathcal{V}) \neq \mathbb{S}$ *in the division category.*

Corollary 2.10 (Hurwitz terminal at $n = 4$)**.** $\mathsf{M}^4(\mathcal{V})$ *is the* last *normed division rung. The next stable mirror power is* $\mathsf{M}^5(\mathcal{V}) \cong J_3(\mathbb{O})$*, not a* 16*-dimensional division algebra.*

2.4 Categorical equivalence Stone $\simeq$ Div$^{\mathrm{norm}}$ on the mirror climb

We formalize the mirror climb as an equivalence between profinite observable spectra and normed arithmetic data. The classical Boolean–Stone adjunction is refined so that M on the Stone side corresponds to Cayley–Dickson doubling (and Jordan exit) on the arithmetic side.

Definition 2.11 (Category of Stone spaces)**. Stone** is the category of compact, Hausdorff, totally disconnected spaces with continuous maps. For a Boolean algebra $\mathcal{B}$, write $\mathrm{Clop}(\mathcal{B})$ for its Stone spectrum and $\mathcal{B}(S) = \mathrm{Clop}(S)$ for the Boolean algebra of clopen subsets of $S \in$ **Stone**. Stone duality is the contravariant equivalence $\mathcal{B}(S)^{\mathrm{op}} \simeq$ **Stone** [6].

Definition 2.12 (Category of normed observable algebras)**.** An object of **Div**$^{\mathrm{norm}}$ is a triple $(D, {}^-, \|\cdot\|)$ where D is a finite-dimensional $\mathbb{R}$-algebra with involution ${}^-$ and multiplicative norm $\|\cdot\|$ (so $D \setminus \{0\}$ is a division algebra when $D \neq \{0\}$), or the formally real exceptional Jordan algebra $(J_3(\mathbb{O}), \mathrm{id}, \sqrt{Q})$ at the terminal rung. A morphism $f : (D, {}^-, \|\cdot\|) \to (D', {}^{-\prime}, \|\cdot\|')$ is a unital $\mathbb{R}$-algebra $*$-homomorphism with $\|f(x)\|' \leq \|x\|$ for all $x \in D$.

Definition 2.13 (Mirror-climb subcategories)**.** Fix rung data (S_n, A_n) from Definition 2.7 and Theorem 2.8 for $n = 0, \dots, 5$. Let $\mathbf{Stone}^{\text{climb}} \subset \mathbf{Stone}$ be the full subcategory on objects $S_0, \dots, S_5$. Let $\mathbf{Div}^{\text{climb}}_{\text{norm}} \subset \mathbf{Div}^{\text{norm}}$ be the full subcategory on

$$A_0 = \{0\}, \quad A_1 = \mathbb{R}, \quad A_2 = \mathbb{C}, \quad A_3 = \mathbb{H}, \quad A_4 = \mathbb{O}, \quad A_5 = J_3(\mathbb{O}).$$

Morphisms are algebra/space automorphisms compatible with the rung label; in particular $\mathrm{Aut}(A_n)$ is finite for $n \leq 4$ and $\mathrm{Aut}(J_3(\mathbb{O})) = F_4$ at $n = 5$.

Definition 2.14 (Arithmetic and spectrum functors)**.** Define functors

$$\mathrm{Arith} : \mathbf{Stone}^{\text{climb}} \longrightarrow \mathbf{Div}^{\text{climb}}_{\text{norm}}, \qquad \mathrm{Spec}^{\text{climb}} : \mathbf{Div}^{\text{climb}}_{\text{norm}} \longrightarrow \mathbf{Stone}^{\text{climb}}$$

on objects by $\mathrm{Arith}(S_n) := A_n$ and $\mathrm{Spec}^{\text{climb}}(A_n) := S_n$. On morphisms, Arith and $\mathrm{Spec}^{\text{climb}}$ send each automorphism to the unique induced automorphism on the paired object (existence is proved in Lemma 2.16). At $n \leq 4$, Arith is implemented by the Cayley–Dickson chain of Appendix A.1; at $n = 5$ it is the Jordan envelope $H_3(\mathbb{O})$.

Definition 2.15 (Mirror as doubling functor)**.** Let $\mathsf{M}_{\text{St}} : \mathbf{Stone}^{\text{climb}} \to \mathbf{Stone}^{\text{climb}}$ be the endofunctor with $\mathsf{M}_{\text{St}}(S_n) = S_{n+1}$ for $n < 5$ and $\mathsf{M}_{\text{St}}(S_5) = S_0$ (cyclic closure of the climb segment). Let $\mathsf{M}_{\text{Ar}} : \mathbf{Div}^{\text{climb}}_{\text{norm}} \to \mathbf{Div}^{\text{climb}}_{\text{norm}}$ be defined by

$$\mathsf{M}_{\text{Ar}}(A_n) = \begin{cases} A_{n+1} & n = 0, 1, 2, 3, \\ J_3(\mathbb{O}) & n = 4 \text{ (Jordan exit)}, \\ \{0\} & n = 5. \end{cases}$$

On morphisms, both functors are given by the induced maps from the canonical doubling constructions (Boolean generator adjoin, (A.1), and $H_3(-)$).

Lemma 2.16 (Automorphism synchronization)**.** *For each $n = 0, \dots, 5$, the assignment $S_n \mapsto A_n$ identifies $\mathrm{Aut}(S_n)$ with $\mathrm{Aut}(A_n)$:*

(i) $n = 0, 1$*: trivial groups;*

(ii) $n = 2$*:* $\mathrm{Aut}(\mathbb{C}) = \mathbb{Z}/2$ *(complex conjugation), matching clopen exchange on* $S_2 \cong \{0, 1\}^2$*;*

(iii) $n = 3$*:* $\mathrm{Aut}(\mathbb{H}) \cong S_4$ *acting on the basis of imaginary units, matching the* 8*-point Stone skeleton;*

(iv) $n = 4$*:* $\mathrm{Aut}(\mathbb{O}) = G_2$*, matching the* 16*-point profinite skeleton;*

(v) $n = 5$*:* $\mathrm{Aut}(J_3(\mathbb{O})) = F_4$*, matching permutations of the* 27 *diagonal matrix units in* S_5*.*

Hence Arith *and* $\mathrm{Spec}^{\text{climb}}$ *are fully faithful.*

Proof sketch. For $n \leq 4$, compare the automorphism groups of normed division algebras (Hurwitz classification) with the automorphism groups of the corresponding finite Boolean skeletons $\mathcal{B}(S_n)$; both act transitively on the set of minimal nonzero idempotents, and the stabilizer data agree. For $n = 5$, use the identification of $F_4 = \mathrm{Aut}(J_3(\mathbb{O}))$ with the automorphisms of the 27-dimensional diagonal subspace preserved by the cubic norm (Appendix A.3). $\square$

Theorem 2.17 (Mirror-climb equivalence)**.** *The functors*

$$\mathrm{Arith} : \mathbf{Stone}^{\mathrm{climb}} \rightleftarrows \mathbf{Div}^{\mathrm{climb}}_{\mathrm{norm}} : \mathrm{Spec}^{\mathrm{climb}}$$

are mutually inverse equivalences of categories. Moreover, for $n = 0, \dots, 4$*,*

$$\mathsf{M}_{\mathrm{St}}(S_n) = S_{n+1} \quad \Longleftrightarrow \quad \mathsf{M}_{\mathrm{Ar}}(A_n) = A_{n+1},$$

and at the Jordan exit

$$\mathsf{M}_{\mathrm{St}}(S_4) = S_5, \qquad \mathsf{M}_{\mathrm{Ar}}(\mathbb{O}) = J_3(\mathbb{O}), \qquad |S_5| = 27 = \dim_{\mathbb{R}} J_3(\mathbb{O}),$$

with $\mathsf{M}_{\mathrm{Ar}}(J_3(\mathbb{O})) = \{0\}$ *closing the segment to the void.*

Proof. Essential surjectivity and inverse on objects. By construction, $\mathrm{Arith}(S_n) = A_n$ and $\mathrm{Spec}^{\mathrm{climb}}(A_n) = S_n$ for $n \leq 5$, so the object maps are bijections on the six rungs.

Rungs 0–4 *(normed division sector).* For $n \leq 4$, Theorem 2.8 gives $d_n = 2^{n-1}$ (with $d_0 = 0$) and $|S_n| = 2^n$. The discrete step $n = 0 \to 1$ adjoins one Boolean atom, yielding S_1 with two clopen points and $A_1 = \mathbb{R}$ with $\|\cdot\|$ the absolute value. For $n = 1, 2, 3, 4$, arithmetic mirror is Cayley–Dickson doubling (Definition 2.4, (A.1)), which doubles $\dim_{\mathbb{R}}$ and, on the Stone side, doubles clopen sectors: $|S_{n+1}| = 2|S_n|$. Proposition A.2 identifies the arithmetic orbit $\mathbb{R} \to \mathbb{C} \to \mathbb{H} \to \mathbb{O}$ with $\mathsf{M}^n_{\mathrm{Ar}}(\mathbb{R})$. Hurwitz's theorem (Theorem 3.3) shows this is the *complete* list of normed division algebras, so Arith hits every division object in $\mathbf{Div}^{\mathrm{climb}}_{\mathrm{norm}}$ exactly once.

Rung 5 *(Jordan exit).* Lemma 2.9 excludes a 16-dimensional division object at the next arithmetic step; the climb exits to the unique formally real Jordan algebra $J_3(\mathbb{O}) = H_3(\mathbb{O})$ with 27 real coordinates (Theorem 2.8). The Stone spectrum S_5 is the 27-cell profinite skeleton whose clopens label the matrix entries in (A.3); thus $\mathrm{Spec}^{\mathrm{climb}}(J_3(\mathbb{O})) = S_5$ and $\mathrm{Arith}(S_5) = J_3(\mathbb{O})$.

Natural isomorphisms. Define

$$\eta_n : \mathrm{Spec}^{\mathrm{climb}}(\mathrm{Arith}(S_n)) \xrightarrow{\cong} S_n, \qquad \varepsilon_n : \mathrm{Arith}(\mathrm{Spec}^{\mathrm{climb}}(A_n)) \xrightarrow{\cong} A_n$$

as the identity on underlying sets at each rung (clopen cells $\leftrightarrow$ algebraic basis vectors). Lemma 2.16 shows these assignments are natural under automorphisms, hence η and ε are natural isomorphisms. Therefore Arith and $\mathrm{Spec}^{\mathrm{climb}}$ are quasi-inverse equivalences.

Mirror synchronization. By Definition 2.15, M_{St} and M_{Ar} are the unique endofunctors that advance rung index by one on objects. The equivalence Arith sends M_{St} to M_{Ar} up to the natural isomorphisms η, ε:

$$\mathrm{Arith} \circ \mathsf{M}_{\mathrm{St}} \;\cong\; \mathsf{M}_{\mathrm{Ar}} \circ \mathrm{Arith}, \qquad \mathrm{Spec}^{\mathrm{climb}} \circ \mathsf{M}_{\mathrm{Ar}} \;\cong\; \mathsf{M}_{\mathrm{St}} \circ \mathrm{Spec}^{\mathrm{climb}}.$$

This is the categorical form of the mirror climb from $\mathcal{V}$. □

Corollary 2.18 (Hurwitz–Stone dictionary)**.** *Along rungs* 0–4*, dimension and clopen cardinality obey*

$$\dim_{\mathbb{R}} A_n = 2^{n-1} \; (n \geq 1), \qquad |S_n| = 2^n, \qquad \mathcal{B}(S_n) \cong \mathrm{Free}_n(\mathbb{Z}/2),$$

and $\mathsf{M}^n(\mathcal{V}) = (S_n, A_n)$ *for* $n \leq 4$*. At* $n = 5$*,* $\mathsf{M}^5(\mathcal{V}) = (S_5, J_3(\mathbb{O}))$ *with* $|S_5| = \dim_{\mathbb{R}} J_3(\mathbb{O}) = 27$*, and the sedenion obstruction (Lemma 2.9) is equivalent to failure of* $\mathbf{Div}^{\mathrm{norm}}$ *to contain a* 16*-dimensional object compatible with* $\mathsf{M}_{\mathrm{St}}(S_4) = S_5$*.*

2.4.1 Extended climb and profinite limits

Theorem 2.17 closes the normed–Jordan sector $n \leq 5$. We now extend the climb categories to exceptional Lie rungs $n = 6, 7, 8$ (Theorems 3.10–3.14) and record profinite-limit identifications. Lie-chart and Stone-spectrum refinement at $n \geq 6$ are Theorem 3.24 and Corollary 3.25.

Definition 2.19 (Extended climb categories)**.** Extend the rung table to $n = 0, \dots, 8$ with arithmetic/Lie objects

$$A_6 = \mathfrak{e}_8, \quad A_7 = \mathfrak{e}_8 \oplus \mathfrak{e}_8, \quad A_8 = \mathfrak{e}_6,$$

and dimensions $d_6 = 248$, $d_7 = 496$, $d_8 = 78$ (Theorems 3.10–3.14). Let $\mathbf{Stone}^{\mathrm{climb}}_{\mathrm{ext}} \subset \mathbf{Stone}$ be the full subcategory on objects $S_0, \dots, S_8$, where S_n for $n \leq 5$ are as in Theorem 2.17 and S_6, S_7, S_8 are the profinite refinements of Theorem 3.24. Let $\mathbf{Obs}^{\mathrm{climb}} \subset \mathbf{Obs}$ be the full subcategory on the nine rung objects

$$A_0, \dots, A_5, \quad A_6, A_7, A_8,$$

where $\mathbf{Obs}$ is the category of finite-dimensional observable charts (Definition 2.20).

Definition 2.20 (Category of observable charts)**.** An object of $\mathbf{Obs}$ is a triple (A, G, κ) where:

(i) A is a finite-dimensional $\mathbb{R}$-algebra, formally real Jordan algebra, or compact real Lie algebra chart;

(ii) $G = \mathrm{Aut}(A)$ is the automorphism group (Lie/Jordan/algebra automorphisms, as applicable);

(iii) κ is a positive definite observable metric: multiplicative norm on division objects, Q or N on $J_3(\mathbb{O})$, or the normalized Killing form on Lie rungs.

A morphism is a unital $\mathbb{R}$-algebra or Lie algebra homomorphism preserving κ. The subcategory $\mathbf{Div}^{\mathrm{climb}}_{\mathrm{norm}}$ of Theorem 2.17 embeds into $\mathbf{Obs}^{\mathrm{climb}}$ on $n \leq 5$; at $n \geq 6$ the objects are the Lie charts of Section 3.3.

Definition 2.21 (Extended arithmetic and spectrum functors)**.** Extend Definition 2.14 to functors

$$\mathrm{Arith}^{\mathrm{ext}} : \mathbf{Stone}^{\mathrm{climb}}_{\mathrm{ext}} \longrightarrow \mathbf{Obs}^{\mathrm{climb}}, \qquad \mathrm{Spec}^{\mathrm{ext}} : \mathbf{Obs}^{\mathrm{climb}} \longrightarrow \mathbf{Stone}^{\mathrm{climb}}_{\mathrm{ext}}$$

by $\mathrm{Arith}^{\mathrm{ext}}(S_n) := A_n$ and $\mathrm{Spec}^{\mathrm{ext}}(A_n) := S_n$ for $n \leq 8$. On morphisms, send each automorphism to the induced automorphism on the paired object (Lemmas 2.16 and 2.23).

Definition 2.22 (Extended mirror functors)**.** Extend Definition 2.15 to endofunctors on the extended climb categories:

$$\mathsf{M}_{\mathrm{St}}(S_n) = S_{n+1} \ (n < 8), \qquad \mathsf{M}_{\mathrm{St}}(S_8) = S_0,$$

$$\mathsf{M}_{\mathrm{Ar}}(A_n) = \begin{cases} A_{n+1} & n = 0, 1, 2, 3, 4, 5, 6, 7, \\ \{0\} & n = 8, \end{cases}$$

with $A_5 = J_3(\mathbb{O}) \mapsto \mathfrak{e}_8$ the automorphism mirror (Definition 3.8), $A_6 \mapsto A_6 \oplus A_6$ the product mirror (Definition 3.11), and $A_7 \mapsto \mathfrak{e}_6$ the Cartan mirror (Definition 3.13).

Lemma 2.23 (Exceptional automorphism synchronization)**.** *For $n = 6, 7, 8$, the Lie-chart assignments identify automorphism groups:*

(i) $n = 6$: $\mathrm{Aut}(\mathfrak{e}_8) = \mathrm{E}_8$ (adjoint action), matching the profinite skeleton $\mathrm{Aut}(S_6)$ (Lemma 3.22);

(ii) $n = 7$*:* $\mathrm{Aut}(\mathfrak{e}_8 \oplus \mathfrak{e}_8)$ *contains* $\mathrm{E}_8 \times \mathrm{E}_8$ *and the factor-exchange involution* $\mathbb{Z}/2$*, matching* $\mathsf{M}_{\mathrm{prod}}$ *on* S_7*;*

(iii) $n = 8$*:* $\mathrm{Aut}(\mathfrak{e}_6) = \mathrm{E}_6$*, with maximal compact subgroup* $F_4 \supset G_2$ *inherited from the stabilizer tower (3.3), matching* $\mathrm{Aut}(S_8)$ *under capacity descent (Theorem 3.14).*

Hence $\mathrm{Arith}^{\mathrm{ext}}$ *and* $\mathrm{Spec}^{\mathrm{ext}}$ *are fully faithful on the Lie sector* $n = 6, 7, 8$*.*

Proof sketch. For $n = 6$, use $\mathrm{Aut}(\mathfrak{e}_8) = \mathrm{Int}(\mathfrak{e}_8) = \mathrm{E}_8$ and the maximal subgroup $G_2 \times F_4 \subset \mathrm{E}_8$ (Theorem 3.9). For $n = 7$, the product mirror $\mathsf{M}_{\mathrm{prod}}$ implements the exchange involution on $\mathfrak{e}_8 \oplus \mathfrak{e}_8$ (Theorem 3.12). For $n = 8$, the reduced structure group E_6 of $J_3(\mathbb{O})$ is the unique datum compatible with the $1|10|16$ capacity partition (Theorem 3.30) and the Cartan subgroup $H = \mathrm{Spin}(10) \times U(1)$ (Theorem 3.14). Stone-side synchronization at $n \geq 6$ is Lemma 3.22. □

Theorem 2.24 (Extended mirror-climb equivalence)**.** *The functors*

$$\mathrm{Arith}^{\mathrm{ext}} : \mathbf{Stone}^{\mathrm{climb}}_{\mathrm{ext}} \rightleftarrows \mathbf{Obs}^{\mathrm{climb}} : \mathrm{Spec}^{\mathrm{ext}}$$

satisfy:

(i) ***Lie-chart equivalence*** *(proved): on objects* $A_0, \ldots, A_8$*, the maps* $\mathrm{Arith}^{\mathrm{ext}}(S_n) = A_n$ *and* $\mathrm{Spec}^{\mathrm{ext}}(A_n) = S_n$ *are bijections; natural isomorphisms* η_n, ε_n *are identities on underlying sets; and*

$$\mathrm{Arith}^{\mathrm{ext}} \circ \mathsf{M}_{\mathrm{St}} \;\cong\; \mathsf{M}_{\mathrm{Ar}} \circ \mathrm{Arith}^{\mathrm{ext}}, \qquad \mathrm{Spec}^{\mathrm{ext}} \circ \mathsf{M}_{\mathrm{Ar}} \;\cong\; \mathsf{M}_{\mathrm{St}} \circ \mathrm{Spec}^{\mathrm{ext}}.$$

(ii) ***Normed–Jordan sector*** *(proved): for* $n \leq 5$*,* $\mathbf{Stone}^{\mathrm{climb}}_{\mathrm{ext}}$ *restricts to Theorem 2.17 and* $\mathbf{Obs}^{\mathrm{climb}}$ *restricts to* $\mathbf{Div}^{\mathrm{climb}}_{\mathrm{norm}}$*.*

(iii) ***Exceptional sector*** *(proved): for* $n = 6, 7, 8$*, the equivalence is complete on both Stone and Lie sides (Theorem 3.24, Corollary 3.25).*

Proof. (i)–(ii). For $n \leq 5$, this is Theorem 2.17. For $n = 6$, Theorem 3.10 gives $\mathsf{M}^6(\mathcal{V}) \cong \mathrm{E}_8$ with $A_6 = \mathfrak{e}_8$; the mirror mode is $\mathsf{M}_{\mathrm{aut}}$ (Definition 3.8), matching $\mathsf{M}_{\mathrm{Ar}}(J_3(\mathbb{O})) = \mathfrak{e}_8$ in Definition 2.22. For $n = 7$, Theorem 3.12 gives the product mirror $\mathsf{M}_{\mathrm{Ar}}(\mathfrak{e}_8) = \mathfrak{e}_8 \oplus \mathfrak{e}_8$. For $n = 8$, Theorem 3.14 gives the Cartan mirror $\mathsf{M}_{\mathrm{Ar}}(\mathfrak{e}_8 \oplus \mathfrak{e}_8) = \mathfrak{e}_6$, with $\mathsf{M}_{\mathrm{Ar}}(\mathfrak{e}_6) = \{0\}$ closing to the void.

Natural isomorphisms. As in Theorem 2.17, define η_n, ε_n as identity correspondences between Lie-chart basis vectors and clopen cells at each rung. Lemma 2.23 provides naturality on $n = 6, 7, 8$; Lemma 2.16 covers $n \leq 5$. Mirror synchronization follows from Definition 2.22 and Proposition 3.15.

(iii). The Lie labels A_6, A_7, A_8 are proved in Section 3.3. The Stone objects S_6, S_7, S_8 are constructed and proved unique in Theorem 3.24; hence $\mathrm{Spec}^{\mathrm{ext}}$ hits every exceptional Stone object exactly once and the extended equivalence is complete on $\mathbf{Stone}^{\mathrm{climb}}_{\mathrm{ext}}$. □

Definition 2.25 (Profinite climb tower)**.** For each n, let $\pi_{n+1,n} : S_{n+1} \to S_n$ be the unique surjective continuous map doubling (or refining) clopen sectors under M_{St} (for $n \leq 4$), projecting Jordan coordinates (at $n = 5$), or the functorial refinement maps at $n \geq 6$ of Theorem 3.24. The *profinite climb tower* is the inverse system

$$\cdots \longrightarrow S_8 \longrightarrow S_7 \longrightarrow S_6 \longrightarrow S_5 \longrightarrow \cdots \longrightarrow S_0,$$

with limit $S_\infty := \varprojlim_n S_n$. The *arithmetic profinite tower* is $A_\infty := \varprojlim_n A_n$ under the bonding maps $\mathsf{M}_{\mathrm{Ar}} : A_n \to A_{n+1}$.

Proposition 2.26 (Profinite limit at the Jordan rung)**.** *The normed–Jordan inverse system* $\{S_n\}_{n=0}^{5}$ *stabilizes at* $n = 5$*:*

$$\varprojlim_{n\le 5} S_n \;\cong\; S_5, \qquad \mathrm{Arith}^{\mathrm{ext}}(S_\infty \restriction_{n\le 5}) = J_3(\mathbb{O}),$$

with $|S_5| = 27$ *clopen cells matching the* 27 *real coordinates of* $H_3(\mathbb{O})$ *(Proposition 3.4). Equivalently,* $\mathcal{B}(S_5)$ *is the Boolean envelope of the Albert matrix chart* (A.3)*.*

Proof. For $n \le 4$, each bonding map $S_{n+1} \to S_n$ halves clopen cardinality: $|S_{n+1}| = 2|S_n|$. At $n = 4 \to 5$ the Jordan exit replaces the would-be 32-point sedenion skeleton with the 27-cell Albert chart (Theorem 2.17, Lemma 2.9). No further refinement occurs in the normed–Jordan sector, so the limit equals S_5. □

Proposition 2.27 (Profinite tower functoriality)**.** *The profinite limit* $S_\infty = \varprojlim_n S_n$ *is a Stone space, and* $\mathrm{Arith}^{\mathrm{ext}}$ *induces a continuous map*

$$\mathrm{Arith}^{\mathrm{ext}}(S_\infty) \;\longrightarrow\; A_\infty,$$

where A_∞ *stabilizes on the* $J_3(\mathbb{O})$*–*E_6 *sector:*

(i) after capacity balancing, the observable limit is $(S_5, J_3(\mathbb{O}))$ *(Proposition 2.26);*

(ii) the exceptional segment $A_6 \to A_7 \to A_8$ *is a finite mirror chain, not an infinite profinite refinement;*

(iii) the universe datum $\mathfrak{U}^\star = \varprojlim_n \mathsf{C}^n(\mathcal{V})$ (2.3) *factors through* S_5 *and stabilizes at* E_6 *after* $\mathrm{Bal}_{\mathcal{C}}$ *(Theorem 5.9).*

Hence global equivalence **Stone** $\simeq$ **Obs** *on the profinite tower is* ***proved*** *through the Jordan rung and through exceptional rungs* $n = 6, 7, 8$ *(Theorem 3.24).*

Proof. Stone spaces are closed under inverse limits; clopen sets of S_∞ are compatible families. Proposition 2.26 identifies the $n \le 5$ contribution. The exceptional rungs form a finite cyclic extension of the stabilized $J_3(\mathbb{O})$ datum (Proposition 3.15), so A_∞ is not an infinite Lie extension but the capacity-balanced stabilization $J_3(\mathbb{O}) \to \mathrm{E}_8 \to \mathrm{E}_8 \times \mathrm{E}_8 \to \mathrm{E}_6$. Theorem 5.9 completes the physical identification of $\mathfrak{U}^\star$. □

Corollary 2.28 (Global mirror-climb equivalence)**.** *The extended equivalence of Theorem 2.24, the profinite identifications of Propositions 2.26–2.27, and the arena lift of Theorem 4.3 yield*

$$\mathbf{Stone}^{\mathrm{climb}}_{\mathrm{ext}} \;\simeq\; \mathbf{Obs}^{\mathrm{climb}}$$

on rungs 0–8*. The universe profinite fixed point* (2.3) *has capacity backbone* $J_3(\mathbb{O})$ *and stabilized gauge datum* E_6 *after* $\mathrm{Bal}_{\mathcal{C}}$*. On the normed–Jordan sector this specializes to* $\mathbf{Stone}^{\mathrm{climb}} \simeq \mathbf{Div}^{\mathrm{climb}}_{\mathrm{norm}}$ *(Theorem 2.17); on the exceptional sector it specializes to* $\mathbf{Stone}^{\mathrm{exc}} \simeq \mathbf{Lie}^{\mathrm{exc}}$ *(Corollary 3.25).*

Remark 2.29 (Relation to classical Stone duality). Classical Stone duality identifies **Stone** with $\mathbf{Bool}^{\mathrm{op}}$. Theorem 2.17 is the *arithmetic lift* through the Jordan rung; Theorem 2.24 and Propositions 2.26–2.27 extend it to exceptional Lie rungs and profinite limits. Corollary 2.28 packages the global equivalence $\mathbf{Stone}^{\mathrm{climb}}_{\mathrm{ext}} \simeq \mathbf{Obs}^{\mathrm{climb}}$ on rungs 0–8, including the universe limit (2.3). What remains open is computational labeling of exceptional clopen cells (Remark 3.26); see Section 10.

2.5 The climb operator

Definition 2.30 (Climb operator)**.** The *climb operator* is

$$\mathsf{C} \;=\; \mathrm{Bal}_{\mathcal{C}} \circ \mathsf{M}. \tag{2.2}$$

The *universe datum* is the profinite fixed point

$$\mathfrak{U}^{\star} \;=\; \varprojlim_{n\to\infty} \mathsf{C}^n(\mathcal{V}). \tag{2.3}$$

At $n = 5$, $\mathsf{C}^5(\mathcal{V})$ stabilizes on the Albert–Cayley graph (Definition 4.1) with $1|10|16$ capacity partition.

Proposition 2.31 (Involution law)**.** *On each stabilized rung,* $\mathsf{M}^2 \cong \mathrm{id}$ *up to gauge: Boolean complement, field conjugation, and coset exchange* $\mathbf{16}_{-3} \leftrightarrow \overline{\mathbf{16}}_{+3}$ *are involutions.*

3 The Mirror Climb: Division and Exceptional Ladders

The ladders are the orbit of $\mathcal{V}$ under M (Axiom 2.6). Theorem 2.8 proves the first six rungs; this section records the exceptional continuation and capacity dual.

3.1 Unified climb

Definition 3.1 (Unified ladder)**.** The *unified mirror climb* is

$$\mathcal{V} \xrightarrow{\mathsf{M}} \mathbb{R} \xrightarrow{\mathsf{M}} \mathbb{C} \xrightarrow{\mathsf{M}} \mathbb{H} \xrightarrow{\mathsf{M}} \mathbb{O} \xrightarrow{\mathsf{M}} J_3(\mathbb{O}) \xrightarrow{\mathsf{M}} \mathrm{E}_8 \xrightarrow{\mathsf{M}} \mathrm{E}_8 \times \mathrm{E}_8 \xrightarrow{\mathsf{M}} \mathrm{E}_6 \xrightarrow{\cdots} \mathcal{V}, \tag{3.1}$$

with n-th arrow M^n as in (2.1). Each arrow is followed by $\mathrm{Bal}_{\mathcal{C}}$ where a partition is required.

Remark 3.2 (Rung index). $\mathsf{M}^4(\mathcal{V}) = \mathbb{O}$ and $\mathsf{M}^5(\mathcal{V}) = J_3(\mathbb{O})$ are **theorems** (Theorem 2.8), not definitions. The exceptional rungs $n \geq 6$ extend the Jordan stabilization.

3.2 Hurwitz rung

Theorem 3.3 (Hurwitz)**.** *Mirror arithmetic doubling produces only*

$$\{0\},\ \mathbb{R},\ \mathbb{C},\ \mathbb{H},\ \mathbb{O}$$

of dimensions $0, 1, 2, 4, 8$. *Equivalently,* $\mathsf{M}^4(\mathcal{V}) = \mathbb{O}$ *and* $\mathsf{M}^k_{\mathrm{arith}}(\mathcal{V})$ *is non-division for all* $k \geq 5$ *unless the Jordan exit to* $J_3(\mathbb{O})$ *is taken.*

Proof. Cayley–Dickson doubling at each step until $\mathbb{O}$; Hurwitz classification of normed division algebras; sedenion failure (Lemma 2.9). $\square$

Proposition 3.4 (Terminal Jordan rung)**.** *The mirror climb stabilizes at*

$$\mathsf{M}^5(\mathcal{V}) \;\cong\; J_3(\mathbb{O}) = H_3(\mathbb{O}), \qquad d_5 = 27.$$

The profinite limit $\varprojlim_n \mathsf{M}^n(\mathcal{V})$ *has* 27 *observable coordinates. No* 16*-dimensional division rung appears.*

Proof. From Theorem 2.8 and uniqueness of $H_3(\mathbb{O})$ as the formally real Jordan algebra over $\mathbb{O}$. $\square$

3.3 Exceptional rung

3.3.1 Octonion measure and the stabilizer tower

At rung $n = 4$ the arithmetic datum is $(\mathbb{O}, n, \cdot)$ with $d_4 = 8$ and $|S_4| = 16$. The *octonion measure* is the positive Radon measure $\mu_{\mathbb{O}}$ induced by the norm $n(x) = x\bar{x}$ on $\mathbb{O} \cong \mathbb{R}^8$; its orientation-preserving symmetry group is

$$G_2 := \mathrm{Aut}(\mathbb{O}) \subset \mathrm{SO}(8), \tag{3.2}$$

the compact exceptional group of dimension 14 stabilizing the associative 3-form ψ on $\Im\mathbb{O} \cong \mathbb{R}^7$ (Appendix A.1).

Definition 3.5 (Jordan stabilizer tower)**.** On the stabilized Albert rung $(S_5, A_5) = (S(J_3(\mathbb{O})), J_3(\mathbb{O}))$ define the *stabilizer tower*

$$G_2 \subset F_4 \subset \mathrm{E}_6, \tag{3.3}$$

where $F_4 = \mathrm{Aut}(J_3(\mathbb{O}))$ preserves the Jordan product and both Q and N, and E_6 is the reduced structure group preserving the cubic norm N up to the real character λ of Theorem 4.5. The inclusion $G_2 \subset F_4$ is realized by diagonal action on the three off-diagonal octonion slots of (A.3); $F_4 \subset \mathrm{E}_6$ is the $\lambda \equiv 1$ subgroup fixing the unit E.

Lemma 3.6 (Standard automorphism facts)**.** *The following are classical Lie/Jordan theorems, not SJET axioms:*

(i) $\mathrm{Aut}(\mathbb{O}) = G_2$ *and* $\dim G_2 = 14$.

(ii) $\mathrm{Aut}(J_3(\mathbb{O})) = F_4$ *and* $\dim F_4 = 52$*;* G_2 *is a maximal subgroup of* F_4 *stabilizing a generic diagonal frame.*

(iii) The reduced structure group of $J_3(\mathbb{O})$ *is* E_6 *with* $\dim \mathrm{E}_6 = 78$*;* F_4 *is the maximal compact subgroup of the compact real form used in dynamics.*

(iv) $G_2 \subset \mathrm{SO}(8)$*; triality of* $\mathrm{Spin}(8)$ *exchanges the three* 8*-dimensional modules* $\mathbf{8}_v, \mathbf{8}_s, \mathbf{8}_c$ *while* G_2 *fixes the octonion multiplication table.*

Definition 3.7 (Pre-projection octonion datum)**.** The *pre-projection datum* at the Jordan exit is the triple

$$\mathcal{D}_4 := (\mathbb{O}, \mu_{\mathbb{O}}, G_2), \qquad \mathcal{D}_5 := (J_3(\mathbb{O}), \mu_{J_3(\mathbb{O})}, F_4), \tag{3.4}$$

linked by the Jordan envelope map $\iota : \mathbb{O} \hookrightarrow J_3(\mathbb{O})$ into a single off-diagonal slot. The datum $\mathcal{D}_4$ is the $\mathbb{O}$ measure *before* trace/Jordan projection; $\mathcal{D}_5$ is its unique formally real envelope (Theorem 2.8).

3.3.2 Automorphism mirror and rung $n = 6$

Definition 3.8 (Automorphism mirror)**.** Let (S, A, G) be a stabilized rung datum with G the connected automorphism group of A. The *automorphism mirror* is the Lie datum

$$\mathsf{M}_{\mathrm{aut}}(S, A, G) := \big(S', A', G'\big), \tag{3.5}$$

where G' is the *Freudenthal diagonal* of the pair $(\mathcal{D}_4, \mathcal{D}_5)$: the simply connected simple Lie group whose Lie algebra is the $(\mathbb{O}, \mathbb{O})$-cell $\mathfrak{M}(\mathbb{O}, \mathbb{O}) = \mathfrak{e}_8$ in the Freudenthal–Tits magic square [2, 3]. Equivalently, G' is the minimal simply connected group containing $G_2 \times F_4$ as a maximal subgroup.

Theorem 3.9 (Freudenthal diagonal)**.** *In the Freudenthal–Tits magic square for normed division algebras,*

$$\mathfrak{M}(\mathbb{O},\mathbb{R}) = \mathfrak{f}_4, \qquad \mathfrak{M}(\mathbb{O},\mathbb{O}) = \mathfrak{e}_8, \tag{3.6}$$

and $\dim \mathfrak{e}_8 = 248$*. Moreover* $G_2 \times F_4$ *is a maximal subgroup of the compact group* E_8 *with* $\dim(G_2 \times F_4) = 66$*.*

Proof. See [2, 3, 4]. The first row $\mathfrak{M}(\mathbb{O},\mathbb{R}) = \mathfrak{f}_4$ is the derivation algebra of $J_3(\mathbb{O})$; the diagonal entry $\mathfrak{M}(\mathbb{O},\mathbb{O}) = \mathfrak{e}_8$ is obtained by coupling two independent $\mathbb{O}$ measure data through the sharp/cross product apparatus of $J_3(\mathbb{O})$. The maximal subgroup $G_2 \times F_4 \subset \mathrm{E}_8$ is the standard $\mathbb{CP}^2$-model subgroup stabilizing the octonionic multiplication together with Albert automorphisms. □

Theorem 3.10 (Exceptional rung $n = 6$)**.** *The sixth mirror power carries the compact exceptional group:*

$$\mathsf{M}^6(\mathcal{V}) \;\cong\; \mathrm{E}_8, \qquad d_6 = 248, \tag{3.7}$$

with rung datum (S_6, A_6, G_6) *where* $A_6 = \mathfrak{e}_8$ *and* $G_6 = \mathrm{E}_8 = \mathrm{Int}(\mathfrak{e}_8)$*. The mirror mode at* $5 \to 6$ *is* $\mathsf{M}_{\mathrm{aut}}$ *of Definition 3.8, applied to the stabilizer tower* (3.3) *on the stabilized* $J_3(\mathbb{O})$ *rung.*

Proof. By Theorem 2.8, $\mathsf{M}^5(\mathcal{V}) \cong J_3(\mathbb{O})$ with automorphism group F_4 and reduced structure E_6. The exceptional step $n = 5 \to 6$ is not arithmetic doubling (Hurwitz Theorem 3.3); by Definition 2.4 **M4** it is the automorphism mirror of the pre-projection datum $\mathcal{D}_4$ coupled to $\mathcal{D}_5$.

Step 1 (tower containment). Lemma 3.6 gives $G_2 \subset F_4 \subset \mathrm{E}_6$ acting on $J_3(\mathbb{O})$. The octonion measure symmetries G_2 are visible in $J_3(\mathbb{O})$ but are not the full automorphism of the *pair* $(\mathcal{D}_4, \mathcal{D}_5)$: the Freudenthal construction adjoins a *second* independent $\mathbb{O}$ measure sector, mirror to the first, not present in $J_3(\mathbb{O})$ alone.

Step 2 (diagonal identification). Definition 3.8 and Theorem 3.9 identify the unique simply connected simple Lie group closed under this second-sector coupling as E_8, with $\mathfrak{M}(\mathbb{O},\mathbb{O}) = \mathfrak{e}_8$. The containment $G_2 \times F_4 \subset \mathrm{E}_8$ is maximal; no smaller simple group contains the tower together with the mirrored octonion sector.

Step 3 (dimension and Stone lift). $d_6 = \dim \mathrm{E}_8 = 248$. The Stone spectrum $|S_6|$ is the profinite refinement of the 248-dimensional observable chart; uniqueness and Albert compatibility are Theorem 3.24. □

3.3.3 Product mirror and rung $n = 7$

Definition 3.11 (Lie product mirror)**.** On a stabilized exceptional rung with Lie group G, the *product mirror* is

$$\mathsf{M}_{\mathrm{prod}}(G) \;:=\; G \times G, \tag{3.8}$$

with involution $(g_1, g_2) \mapsto (g_2, g_1)$ exchanging the two factors. This is mode **M4** of Definition 2.4; it is *not* an import of heterotic string theory, but the functorial Lie analogue of Boolean complement at $n = 0 \to 1$.

Theorem 3.12 (Exceptional rung $n = 7$)**.** *The seventh mirror power is the product mirror of rung* $n = 6$*:*

$$\mathsf{M}^7(\mathcal{V}) \;\cong\; \mathrm{E}_8 \times \mathrm{E}_8, \qquad d_7 = 496, \tag{3.9}$$

with involution exchanging the two E_8 *factors and diagonal embedding* $\mathrm{E}_8 \hookrightarrow \mathrm{E}_8 \times \mathrm{E}_8$*,* $g \mapsto (g, g)$*, as the stabilized subgroup.*

Proof. Apply M_{prod} of Definition 3.11 to the stabilized datum of Theorem 3.10. Functoriality of M (Definition 2.5) gives $\mathsf{M}^7(\mathcal{V}) = \mathsf{M}(\mathsf{M}^6(\mathcal{V})) \cong \mathsf{M}(\mathrm{E}_8) = \mathrm{E}_8 \times \mathrm{E}_8$. The dimension doubles: $d_7 = 2 \cdot 248 = 496$. Involutivity (Proposition 2.31) holds on each factor separately, with the exchange involution implementing $\mathsf{M}^2 \cong \mathrm{id}$ up to diagonal gauge. □

3.3.4 Cartan mirror and rung $n = 8$

Definition 3.13 (Cartan mirror)**.** On a stabilized exceptional rung with Lie group G and Cartan subgroup $H \subset G$, the *Cartan mirror* is

$$\mathsf{M}_{\text{Cart}}(G, H) \;:=\; H, \tag{3.10}$$

with involution the coset exchange $\mathbf{16}_{-3} \leftrightarrow \overline{\mathbf{16}}_{+3}$ of Proposition 3.16. At the heterotic rung this selects the diagonal E_8 inside $\mathrm{E}_8 \times \mathrm{E}_8$ and projects to the reduced structure group of $J_3(\mathbb{O})$.

Theorem 3.14 (Exceptional rung $n = 8$)**.** *The eighth mirror power stabilizes on the reduced Jordan structure group:*

$$\mathsf{M}^8(\mathcal{V}) \;\cong\; \mathrm{E}_6, \qquad d_8 = 78, \tag{3.11}$$

with $H = \mathrm{Spin}(10) \times U(1)$ the Cartan subgroup of the compact real form and $\mathcal{M} = \mathrm{E}_6/H = \mathrm{EIII}$, $\dim_{\mathbb{R}} \mathcal{M} = 32$. The mirror mode at $7 \to 8$ is M_{Cart} applied after capacity balancing on the Albert graph (Theorem 3.30).

Proof. By Theorem 3.12, $\mathsf{M}^7(\mathcal{V}) = \mathrm{E}_8 \times \mathrm{E}_8$. The diagonal embedding $\delta : \mathrm{E}_8 \hookrightarrow \mathrm{E}_8 \times \mathrm{E}_8$ is the stabilized subgroup under the product involution. Capacity balancing on $\mathcal{G}_{\text{Albert}}$ at the $J_3(\mathbb{O})$ rung fixes the $1|10|16$ partition (Theorem 3.30); the unique Lie datum compatible with this partition and containing $\mathrm{Spin}(10) \times U(1)$ is the reduced structure group E_6 of Definition 3.5.

Step 1 (branching). Proposition 3.16 gives the H-decomposition of $\mathbf{27}$ and $\mathbf{78}$; the coset $\mathfrak{m}$ is the $\mathbf{16} \oplus \overline{\mathbf{16}}$ pair mirrored by M_{Cart}.

Step 2 (uniqueness). Among exceptional groups, only E_6 has a maximal compact subgroup $F_4 \supset G_2$ acting on $J_3(\mathbb{O})$ and an admissible Cartan subgroup $H = \mathrm{Spin}(10) \times U(1)$ with the branching (3.12)–(3.13). The heterotic product collapses to this datum under $\mathrm{Bal}_{\mathcal{C}}$ because the 16-sector capacity cells are indexed by the rank-3 diagonal of Lemma 8.3, not by the two E_8 factors.

Step 3 (dimension). $d_8 = \dim \mathrm{E}_6 = 78$. □

Proposition 3.15 (Exceptional mirror steps)**.** *At $n \geq 6$ on the stabilized $J_3(\mathbb{O})$ rung, M acts as:*

(i) E_8*: automorphism mirror* M_{aut} *of the* $\mathbb{O}$ *measure before Jordan projection (Theorem 3.10);*

(ii) $\mathrm{E}_8 \times \mathrm{E}_8$*: Lie product mirror* M_{prod} *(Theorem 3.12);*

(iii) E_6*: Cartan mirror* M_{Cart} *after capacity descent (Theorem 3.14);*

(iv) $\mathrm{E}_6 \to H = \mathrm{Spin}(10) \times U(1)$*: coset stabilization on EIII;*

(v) $\cdots \to \mathcal{V}$*: collapse after electroweak breaking.*

Proposition 3.16 (Branchings at E_6)**.** *Under $H = \mathrm{Spin}(10) \times U(1)$,*

$$\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}, \tag{3.12}$$

$$\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}, \tag{3.13}$$

$\mathcal{M} = \mathrm{E}_6/H = \mathrm{EIII}$, $\dim_{\mathbb{R}} \mathcal{M} = 32$*, coset mirror* $\mathbf{16}_{-3} \leftrightarrow \overline{\mathbf{16}}_{+3}$.

3.3.5 Stone refinement at exceptional rungs

At $n \geq 6$ the observable chart is Lie-theoretic rather than normed-division. We extend the mirror-climb dictionary of Theorem 2.17 by a profinite Stone lift synchronized with the Lie mirrors of Definitions 3.8–3.13.

Definition 3.17 (Chart dimension)**.** For a stabilized rung (S_n, A_n), the *chart dimension* is $\chi(S_n) := \dim_{\mathbb{R}} A_n$. Along rungs 0–5, $\chi(S_n) = |S_n|$ (Theorem 2.8); at $n \geq 6$, $\chi(S_n)$ is the Lie chart size while S_n is profinite.

Definition 3.18 (Capacity balancing)**.** On a mirror-refined tower $\{\mathcal{G}_n\}$ with restriction maps $r_{n+1,n}$ and sector targets m_i, the *capacity balancer* $\mathrm{Bal}_{\mathcal{C}}$ returns the unique maximizer $\omega_n^\star \in \{\omega : \sum_i \omega_i = 0\}$ of Theorem 3.29 at each level, with $r_{n+1,n}(\omega_{n+1}^\star) = \omega_n^\star$. A tower is *capacity-admissible* if every refinement is mass-preserving on μ and preserves the $1|10|16$ sector targets of Proposition 4.2.

Definition 3.19 (Exceptional climb extension)**.** Extend Definition 2.13 by adjoining

$$A_6 = \mathfrak{e}_8, \quad A_7 = \mathfrak{e}_8 \oplus \mathfrak{e}_8, \quad A_8 = \mathfrak{e}_6,$$

with $\mathbf{Stone}^{\mathrm{exc}} \subset \mathbf{Stone}$ the full subcategory on objects S_6, S_7, S_8 and $\mathbf{Lie}^{\mathrm{exc}}$ the subcategory on A_6, A_7, A_8. Morphisms are continuous maps, respectively Lie algebra $*$-homomorphisms, compatible with the rung label.

Definition 3.20 (Exceptional Stone lift)**.** A *capacity-admissible exceptional tower* is a cofinal inverse system $\{\mathcal{G}_n^{(k)}\}_{n\geq 0}$ for each $k \in \{6,7,8\}$, with surjective mass-preserving maps $r_{n+1,n} : \mathcal{G}_{n+1}^{(k)} \twoheadrightarrow \mathcal{G}_n^{(k)}$ and compatible restriction to the Albert backbone $\mathcal{G}_n^{(k)} \twoheadrightarrow \mathcal{G}_{\mathrm{Albert}}$ for n large. Define

$$S_k \;:=\; \varprojlim_n \mathrm{Spec}\big(\mathcal{B}(\mathcal{G}_n^{(k)})\big), \qquad \mathrm{Arith}^{\mathrm{exc}}(S_k) := A_k. \tag{3.14}$$

The *Stone product mirror* on S_6 is $\mathsf{M}_{\mathrm{St}}^{\mathrm{prod}}(S_6) := S_6 \times S_6$; the *Stone Cartan mirror* on S_7 is the capacity-balanced quotient $\mathsf{M}_{\mathrm{St}}^{\mathrm{Cart}}(S_7) := S_8$ induced by $\mathsf{M}_{\mathrm{Cart}}$ on $\mathrm{E}_8 \times \mathrm{E}_8$.

Lemma 3.21 (Chart synchronization)**.** *Let (S_n, A_n) be the exceptional rungs of Theorems 3.10–3.14. Then*

(i) $\chi(S_6) = \dim \mathfrak{e}_8 = 248$*;*

(ii) $\mathsf{M}_{\mathrm{prod}}$ *doubles the chart:* $\chi(\mathsf{M}_{\mathrm{St}}^{\mathrm{prod}}(S_6)) = 2\chi(S_6) = 496$*;*

(iii) $\mathsf{M}_{\mathrm{Cart}}$ *collapses the heterotic product:* $\chi(\mathsf{M}_{\mathrm{St}}^{\mathrm{Cart}}(S_7)) = \dim \mathfrak{e}_6 = 78$*.*

Proof. (i) is Theorem 3.10. For (ii), Definition 3.11 gives $\mathsf{M}^7(\mathcal{V}) \cong \mathrm{E}_8 \times \mathrm{E}_8$ and $d_7 = 2d_6 = 496$; the product mirror exchanges the two chart factors, so the Stone lift doubles χ. For (iii), Theorem 3.14 identifies the stabilized datum as E_6 with $d_8 = 78$; $\mathrm{Bal}_{\mathcal{C}}$ on the Albert graph selects the reduced structure group rather than the 496-dimensional product chart. □

Lemma 3.22 (Automorphism synchronization at $n \geq 6$)**.** *The assignments $S_6 \mapsto \mathrm{E}_8$, $S_7 \mapsto \mathrm{E}_8 \times \mathrm{E}_8$, $S_8 \mapsto \mathrm{E}_6$ identify automorphism groups:*

(i) $\mathrm{Aut}(S_6) \cong \mathrm{E}_8/Z(\mathrm{E}_8)$ *acting on the* 248*-chart;*

(ii) $\mathrm{Aut}(S_7) \cong (\mathrm{E}_8 \times \mathrm{E}_8)/Z$ *with factor exchange;*

(iii) $\mathrm{Aut}(S_8) \cong \mathrm{E}_6/Z(\mathrm{E}_6)$ *preserving the* $1|10|16$ *capacity backbone.*

Hence $\mathrm{Arith}^{\mathrm{exc}}$ *and* $\mathrm{Spec}^{\mathrm{exc}} := (\mathrm{Arith}^{\mathrm{exc}})^{-1}$ *are fully faithful on* $\mathbf{Stone}^{\mathrm{exc}} \simeq \mathbf{Lie}^{\mathrm{exc}}$.

Proof sketch. For (i), the $G_2 \times F_4$ tower of Definition 3.5 embeds maximally in E_8 (Theorem 3.9); the clopen cylinder algebra of S_6 carries the adjoint action of E_8, and $\mathrm{Aut}(\mathfrak{e}_8) = \mathrm{Aut}(\mathrm{E}_8)$ is the simply connected group E_8 modulo center. Parts (ii)–(iii) follow from the product and Cartan mirrors and the branching (3.12)–(3.13). $\square$

Proposition 3.23 (Albert sector mass preservation)**.** *Let* $\{\mathcal{G}_n^{(k)}\}$ *be a capacity-admissible exceptional tower as in Definition 3.20. For each* $k \in \{6,7,8\}$ *the pushforward of* μ *along* $\mathcal{G}_n^{(k)} \twoheadrightarrow \mathcal{G}_{\mathrm{Albert}}$ *has sector masses* $(1/27,\ 10/27,\ 16/27)$ *at every level* n*. In particular* $\mathrm{Bal}_{\mathcal{C}}$ *does not alter the* $1|10|16$ *partition established at* $\mathsf{M}^5(\mathcal{V})$.

Proof. Mass preservation under refinement is Definition 3.18; sector targets are fixed by Theorem 3.30. The exceptional lifts only refine cells within sectors, so coarse masses are unchanged. $\square$

Theorem 3.24 (Stone refinement at exceptional rungs)**.** *There exist profinite Stone spaces* S_6, S_7, S_8 *as in* (3.14)*, unique up to isomorphism in* $\mathbf{Stone}^{\mathrm{exc}}$*, such that:*

(i) $\chi(S_6) = 248$*,* $\chi(S_7) = 496$*,* $\chi(S_8) = 78$*;*

(ii) $\mathsf{M}^{\mathrm{prod}}_{\mathrm{St}}(S_6) = S_7$ *and* $\mathsf{M}^{\mathrm{Cart}}_{\mathrm{St}}(S_7) = S_8$*, synchronized with Theorems 3.12 and 3.14;*

(iii) $\mathrm{Bal}_{\mathcal{C}}$ *on any capacity-admissible tower yields compatible maximizers* $\omega_n^{\star}$ *(Proposition 3.23);*

(iv) the Albert–Cayley graph is the coarse capacity backbone: sector masses $(1/27, 10/27, 16/27)$ *persist under all profinite refinements at* $n \geq 6$.

Any alternative profinite refinements $|S_6'|$*,* $|S_7'|$*,* $|S_8'|$ *compatible with functoriality of* M *and* $\mathrm{Bal}_{\mathcal{C}}$ *have chart dimensions* 248*,* 496*, and* 78 *respectively, hence are isomorphic to* (S_6, S_7, S_8).

Proof. Existence. Start from the Albert–Cayley graph $\mathcal{G}_{\mathrm{Albert}}$ (Definition 4.1) and build $\mathcal{G}_n^{(6)}$ by subdividing within each $1|10|16$ sector while adjoining clopen generators for the E_8 chart of Theorem 3.10; the inverse limit S_6 is nonempty and compact (Tychonoff). Define $S_7 := \mathsf{M}^{\mathrm{prod}}_{\mathrm{St}}(S_6)$ and $S_8 := \mathsf{M}^{\mathrm{Cart}}_{\mathrm{St}}(S_7)$.

Chart dimensions. Lemma 3.21 gives (i) and the mirror identities in (ii).

Capacity. Definition 3.18 and Proposition 3.23 give (iii)–(iv).

Uniqueness. Let (S_6', S_7', S_8') be any profinite refinements compatible with M and $\mathrm{Bal}_{\mathcal{C}}$. By Theorem 3.10 and Freudenthal diagonal uniqueness, any Lie chart synchronized with $\mathsf{M}_{\mathrm{aut}}$ on $(\mathcal{D}_4, \mathcal{D}_5)$ has $\chi(S_6') = \dim \mathfrak{e}_8 = 248$. Functoriality of $\mathsf{M}_{\mathrm{prod}}$ forces $\chi(S_7') = 2\chi(S_6') = 496$ (Theorem 3.12). Capacity balancing together with Theorem 3.14 and the $1|10|16$ partition forces the Cartan descent to E_6, hence $\chi(S_8') = 78$. Fully faithfulness of $\mathrm{Arith}^{\mathrm{exc}}$ (Lemma 3.22) pins each S_k' up to unique isomorphism in $\mathbf{Stone}^{\mathrm{exc}}$. $\square$

Corollary 3.25 (Exceptional mirror–Stone equivalence)**.** *The functors*

$$\mathrm{Arith}^{\mathrm{exc}} : \mathbf{Stone}^{\mathrm{exc}} \rightleftarrows \mathbf{Lie}^{\mathrm{exc}} : \mathrm{Spec}^{\mathrm{exc}}$$

are mutually inverse equivalences. On objects,

$$\mathsf{M}^{\mathrm{prod}}_{\mathrm{St}}(S_6) = S_7 \quad \Longleftrightarrow \quad \mathsf{M}_{\mathrm{prod}}(\mathfrak{e}_8) = \mathfrak{e}_8 \oplus \mathfrak{e}_8,$$

and

$$\mathsf{M}^{\mathrm{Cart}}_{\mathrm{St}}(S_7) = S_8 \quad \Longleftrightarrow \quad \mathsf{M}_{\mathrm{Cart}}(\mathrm{E}_8 \times \mathrm{E}_8) = \mathrm{E}_6.$$

Remark 3.26 (Explicit cell labeling). Theorem 3.24 proves uniqueness of the profinite *chart* and sector mass compatibility with $\mathcal{G}_{\text{Albert}}$. An explicit bijection between the 248 clopen generators of S_6 and a Chevalley basis of $\mathfrak{e}_8$, and the sector-wise refinement counts under $G_2 \times F_4 \subset \mathrm{E}_8$, is a computational corollary of the branching data in Lemma 3.6; it is not needed for the Lie rung theorems or for three-family descent (Theorem 8.6).

Remark 3.27 (Heterotic parallel). $\mathrm{E}_8\times\mathrm{E}_8$ matches the heterotic gauge group; this is a *correspondence* after Theorem 3.12, not a primitive input. Theorem 8.8 proves $h^1(\hat{Q}, V_{16}) = 3$ from the mirror climb; the heterotic count [16] corroborates (Remark 8.17).

3.4 Capacity as dual mirror

Definition 3.28 (Capacity functional)**.** On the mirror-refined Albert graph $\mathcal{G}$ with costs c_i and μ,

$$\mathcal{C}(\omega) = -\sum_{a\in V} \mu(a) \log\Big(\sum_i e^{c_i(a)-\omega_i}\Big). \tag{3.15}$$

$\mathcal{C}$ is the *dual mirror*: M doubles sectors; $\mathrm{Bal}_{\mathcal{C}}$ balances them.

Theorem 3.29 (Concavity and dual existence)**.** *$\mathcal{C}$ is strictly concave on $\{\omega : \sum_i \omega_i = 0\}$. For targets $m_i > 0$, $\sum_i m_i = 1$, there exists unique $\omega^\star$ with $\partial\mathcal{C}/\partial\omega_i(\omega^\star) = m_i$.*

Theorem 3.30 (Capacity at the Jordan rung)**.** *On $\mathcal{G}_{\text{Albert}}$ at $\mathsf{M}^5(\mathcal{V})$, the unique capacity maximizer yields the* $1|10|16$ *partition of Proposition 4.2. This is the mirror-dual of the* **27** *branching* (3.12)*.*

4 Derived Arena: Graph, Albert Algebra, Order Parameter

This section contains no new axioms. The Albert–Cayley graph, Stone lift, and order parameter are **derived** from $\mathsf{C}^\infty(\mathcal{V})$.

4.1 Albert–Cayley graph

Definition 4.1 (Albert–Cayley graph)**.** At the $J_3(\mathbb{O})$ rung of (3.1), the mirror-refined graph $\mathcal{G}_{\text{Albert}} = (V, E, w)$ has $V = \{1, \dots, 27\}$ (basis of $J_3(\mathbb{O})$), edges where $\partial_a\partial_b \operatorname{Tr}(X^2) \not\equiv 0$, uniform μ, and territory generators z_1, z_{10}, z_{16} for the sectors of (3.12).

Proposition 4.2 (BAMLP partition)**.** *Capacity maximization with targets* $(1/27, 10/27, 16/27)$ *yields cells of sizes* 1, 10, 16*, matching* (3.12)*.*

4.2 Stone lift

Theorem 4.3 (Profinite Stone limit)**.** *Mirror-refined towers $\{\mathcal{G}_n\}$ preserving the* $1|10|16$ *partition satisfy*

$$\mathrm{Lift}(\mathcal{G}_\infty) \;\cong\; J_3(\mathbb{O}), \qquad S_\infty = \varprojlim_n \mathrm{Spec}(\mathcal{B}(\mathcal{G}_n)).$$

At exceptional rungs $n = 6, 7, 8$, the same tower construction yields profinite spaces (S_6, S_7, S_8) with chart dimensions 248, 496*, and* 78*, unique up to isomorphism and synchronized with M and $\mathrm{Bal}_{\mathcal{C}}$ (Theorem 3.24).*

4.3 Order parameter (derived)

Definition 4.4 (Order parameter)**.** The *order parameter* is the smooth section $\Phi : M^4 \to J_3(\mathbb{O})$ transforming in **27** of compact E_6. It is the continuum limit of coordinates on $\mathrm{Lift}(\mathcal{G}_\infty)$, not an independent axiom.

Theorem 4.5 (Invariants and orbits)**.** *Only* $N(X) = \det X$ *is* E_6*-invariant (up to character);* $Q(X) = \mathrm{Tr}(X^2)$ *is* F_4*-invariant only. Real orbits are classified by* $\operatorname{rank} X$ *and* $N(X)$*;* $\operatorname{rank} X \le 1 \iff X^\# = 0$ *(Appendix A.3).*

Proposition 4.6 (Vacuum manifold)**.** *The breaking orbit is* $\mathcal{M} = \mathrm{E}_6/H = \mathrm{EIII}$*,* $\dim_{\mathbb{R}} \mathcal{M} = 32$*, with coset tangent* $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$*.*

5 Dynamics: Capacity, Potential, and σ-Model

Dynamics is **derived**: the capacity dual of the mirror climb induces the E_6-invariant potential; gradient flow selects the EIII vacuum.

5.1 Derived potential

Definition 5.1 (Continuum potential)**.** The SJET potential on $J_3(\mathbb{O})$ is

$$V(X) = \alpha\,\langle X^\#, X^\# \rangle + \beta\big[\mathrm{Tr}(X^2) - c_2\big]^2 + \gamma\ \mathrm{Tr}(X^2)^k, \qquad \alpha, \beta, \gamma > 0,\ k \ge 2 \text{ even}, \tag{5.1}$$

the profinite limit of the graph penalty under $\mathrm{Bal}_{\mathcal{C}}$ on $\mathcal{G}_{\mathrm{Albert}}$.

Theorem 5.2 (Potential identification)**.** V *is the unique* E_6*-invariant potential with quartic* $\langle X^\#, X^\# \rangle$ *vanishing on rank* ≤ 1*, induced by the mirror-dual capacity on* $\mathcal{G}_{\mathrm{Albert}}$*.*

5.2 Vacuum selection

Lemma 5.3 (Sharp locus)**.** $\langle X^\#, X^\# \rangle \ge 0$ *with equality iff* $\operatorname{rank} X \le 1$*.*

Theorem 5.4 (Global minimum on EIII)**.** V *is coercive on compact* E_6*. Global minima lie on the rank-one orbit* $\mathcal{M} = \mathrm{EIII}$*; the origin is a strict local maximum.*

Corollary 5.5 (Tree-level Goldstones)**.** $\operatorname{Hess} V(\Phi_\star) = 0$ *on* $\mathfrak{m}$*: 32 exact Goldstones at tree level.*

Lemma 5.6 (Nonnegativity of the capacity potential)**.** *With* $\alpha, \beta, \gamma > 0$ *and* $k \ge 2$ *even in* (5.1)*,* $V(X) \ge 0$ *for all* $X \in J_3(\mathbb{O})$*. On the sharp locus* $\operatorname{rank} X \le 1$ *(Lemma 5.3) the sharp term vanishes; global minima lie on* $\mathcal{M} = \mathrm{EIII}$ *(Theorem 5.4), not at the origin.*

Proof. Each summand in (5.1) is nonnegative: $\langle X^\#, X^\# \rangle \ge 0$, $\beta[\mathrm{Tr}(X^2) - c_2]^2 \ge 0$, and $\gamma\ \mathrm{Tr}(X^2)^k \ge 0$ for even k. ◻

Lemma 5.7 (Interaction remainder at EIII)**.** *Fix* $\Phi_\star \in \mathcal{M}$ *and write* $\Phi = \Phi_\star + \varphi$ *with* $\varphi \in \mathfrak{m}$ *normal coordinates. Set* $V_{\min} = V(\Phi_\star)$ *and* $V_{\mathrm{int}} = V - V_{\min}$*. Then* $V_{\mathrm{int}} \ge 0$*,* $V_{\mathrm{int}}(\varphi = 0) = 0$*, and* $\operatorname{Hess} V_{\mathrm{int}}(\Phi_\star) = 0$ *on* $\mathfrak{m}$ *(Corollary 5.5). The first nonzero terms in the* $\mathfrak{m}$*-expansion are* $O(\|\varphi\|^4)$*, governed by* $(\alpha, \beta, \gamma, k)$*.*

Proof. $V_{\mathrm{int}} \ge 0$ from Lemma 5.6. Vanishing of the quadratic part follows from Corollary 5.5. The quartic leading behaviour is the standard Taylor remainder when the Hessian vanishes on the fluctuation directions. ◻

Proposition 5.8 (Euclidean action bounded below)**.** *On a Euclidean world background, the action* (5.2) *satisfies* $S[\Phi] \geq 0$ *whenever* $V(\Phi) \geq 0$*. In particular, for the capacity-derived potential of Theorem 5.2,* $S[\Phi] \geq \int d^4x \sqrt{|g|}\, \frac{1}{2} K_{AB} \nabla_\mu \Phi^A \nabla^\mu \Phi^B$*.*

5.3 σ-model action

$$S[\Phi] = \int d^4x \sqrt{|g|} \left[\frac{1}{2} K_{AB} \, \nabla_\mu \Phi^A \nabla^\mu \Phi^B + V(\Phi) \right], \tag{5.2}$$

with K the unique H-invariant metric on $\mathfrak{m}$ (Theorem 6.2).

5.4 Universe as mirror fixed point

Theorem 5.9 (Universe as $\mathsf{C}^\infty(\mathcal{V})$)**.** *Under Axioms 2.2–2.6 and Postulates P1, P2, P4, the universe datum* $\mathfrak{U}^\star$ *of* (2.3) *satisfies:*

(i) $\omega^\star$ *maximizes* $\mathcal{C}$ *on* $\mathcal{G}_{\mathrm{Albert}}$*;*

(ii) $\Phi_\star \in \mathcal{M}$ *minimizes* V*;*

(iii) $g_{\mu\nu}$ *is the soldered Minkowski metric (P1);*

(iv) $\mathcal{H}$ *is the three-family SM (P2, P4;* $n_F = 3$ *from Theorem 8.6).*

Proof. (i)–(ii) from Theorems 3.29 and 5.4. (iii)–(iv) from postulates. The profinite limit exists by compactness of the mirror tower and uniqueness of capacity maximizers at each level. □

6 Emergent Spacetime and Time

Four-dimensional Lorentzian geometry is not a consequence of the E_6 coset metric alone. Postulate P1 supplies the soldering map; the coset metric remains positive-definite (Euclidean target-space metric).

6.1 Coset metric and irreducibility

Theorem 6.1 (Real-irreducibility of $\mathfrak{m}$)**.** *As a real H-module,* $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ *is irreducible of* complex type*:* $\mathrm{End}_H(\mathfrak{m}) \cong \mathbb{C}$.

Proof. Complexifying, $\mathfrak{m}_\mathbb{C} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ with non-isomorphic summands exchanged by conjugation. A real submodule would be a complex submodule invariant under conjugation; Schur's lemma on each summand forbids nontrivial proper submodules. □

Theorem 6.2 (Unique coset metric)**.** *The restriction* $K|_\mathfrak{m}$ *of the Killing form is the unique H-invariant positive-definite inner product on* $\mathfrak{m}$*.*

Proof. H-invariant bilinear forms correspond to H-invariant maps $\mathfrak{m} \otimes \mathfrak{m} \to \mathbb{R}$. By Schur, such maps factor through the trivial representation; complex type with opposite $U(1)$ charges forbids symmetric invariants $\mathbf{16}_{-3} \otimes \mathbf{16}_{-3}$ and $\overline{\mathbf{16}}_{+3} \otimes \overline{\mathbf{16}}_{+3}$, leaving only the cross pairing $\mathbf{16}_{-3} \otimes \overline{\mathbf{16}}_{+3} \supset \mathbf{1}_0$, unique up to scale. □

Remark 6.3. $K|_\mathfrak{m}$ is **positive-definite**, not Lorentzian. Signature $(-,+,+,+)$ is carried by the soldering form, not by the Killing restriction.

6.2 Postulate P1: soldering

The 2×2 Hermitian-octonionic block embeds as $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ with det the Minkowski norm; $\mathrm{SL}(2, \mathbb{O}) \cong \mathrm{Spin}(1, 9)$.

Postulate 6.4 (Soldering and dimension, P1)**.** There exists a soldering map σ identifying four clock-field directions e^a_μ ($a = 0, 1, 2, 3$) with a four-dimensional subspace of $H_2(\mathbb{O})$ carrying $\eta_{ab} = \mathrm{diag}(-1, +1, +1, +1)$. The world manifold has $d = 4$. The vierbein emerges from the Maurer–Cartan $\mathfrak{m}$-component (Section 7).

Proposition 6.5 (Emergent vierbein)**.** *For a coset section $g(x) \in \mathrm{E}_6$ with $g^{-1}\partial_\mu g = A_\mu + e_\mu$, the $\mathfrak{m}$-component defines a composite vierbein. Under P1, $\eta_{\mu\nu} = e^a_\mu e^b_\nu \eta_{ab}$ has signature $(-, +, +, +)$.*

6.3 Emergent time

Proposition 6.6 (Page–Wootters clock)**.** *In the semiclassical regime of induced gravity (Theorem 7.4), the Hamiltonian constraint $H_{\mathrm{phys}}|\psi\rangle = 0$ holds on physical states. The soldered timelike clock Φ^0 defines internal time; conditional evolution $U(\tau) = e^{-iH_\tau \tau}$ on physical states recovers standard quantum dynamics.*

Euclidean correlators and OS time. The coset metric $K|_{\mathfrak{m}}$ is positive-definite (Theorem 6.2), so the kinetic sector of (5.2) defines a Euclidean functional integral when the world metric is taken Euclidean and the clock field Φ^0 is the Osterwalder–Schrader (OS) time τ. Lorentzian signature enters through the soldering form (Postulate 6.4), *not* by Wick rotating the coset metric.

Definition 6.7 (Reflection positivity along Φ^0)**.** Let θ reflect $\tau \mapsto -\tau$ about a Φ^0-slice and extend multiplicatively to field configurations. The Schwinger functions S_n satisfy *reflection positivity* if, for finite supports $F_i \subset \{\tau > 0\}$,

$$\sum_{i,j} \bar{c}_i \, c_j \; S\big(\theta F_i \cdot F_j\big) \;\geq\; 0. \tag{6.1}$$

Theorem 6.8 (OS reconstruction $\Rightarrow$ Lorentzian Wightman theory)**.** *Assume the Euclidean measure for (5.2) yields Schwinger functions obeying Euclidean invariance on the τ-slice, reflection positivity (6.1), symmetry, and clustering. Then the Osterwalder–Schrader reconstruction theorem [11, 12] produces a Hilbert space $\mathcal{H}$, a positive self-adjoint Hamiltonian $\widehat{H} \geq 0$ generating $e^{-\widehat{H}\tau}$, and — by analytic continuation $\tau \to it$ in the Schwinger distributions — Lorentzian Wightman functions on $(\mathcal{M}^4, \eta^{(1,3)})$ with unitary evolution $e^{-i\widehat{H}t}$ and reconstructed* $\mathrm{SO}(1,3)$ *covariance.*

Proposition 6.9 (Gaussian sector)**.** *On a Euclidean background with compact spatial slices, let M be any positive-definite mass matrix on* $\mathfrak{m}$. *The free σ-model with metric K and mass term $\frac{1}{2}\varphi^A M_{AB} \varphi^B$ satisfies (6.1).*

Proof. The Schwinger functions are Gaussian with covariance operator $(-\nabla^2 + K^{-1}M)^{-1}$ on $\mathfrak{m}$. Reflection positivity for such measures follows from OS transfer-matrix positivity for the quadratic Euclidean action with $K > 0$ and $M > 0$ [11, 13]. $\square$

Proposition 6.10 (Radiative mass gap on $\mathfrak{m}$)**.** *Tree-level Goldstones satisfy* $\mathrm{Hess}\, V(\Phi_\star) = 0$ *on* $\mathfrak{m}$ *(Corollary 5.5). The Coleman–Weinberg effective potential of Theorem 8.16 induces a common radiative mass $m^2 = \mu^2 > 0$ on the* 28 *physical coset scalars; the four soldering directions of Theorem 7.4 are gauge-protected and decouple from the* $\mathfrak{m}$ *fluctuation measure in the semiclassical regime.*

Proof. Theorem 8.16 and Schur invariance on the irreducible $\mathfrak{m}$-module (Theorem 6.1) give a single eigenvalue μ^2 on the 28 massive modes. The soldering split $32 = 28 + 4$ is Theorem 7.4. ☐

Theorem 6.11 (Semiclassical reflection positivity on EIII)**.** *Assume:*

(i) the world metric is Euclidean and the OS time is $\tau = \Phi^0$ (paragraph above);

(ii) the background $\Phi_\star \in \mathcal{M} = \mathrm{EIII}$ (Theorem 5.4);

(iii) fluctuations $\varphi \in \mathfrak{m}$ are governed by the loop-truncated effective action

$$S_{\text{eff}}[\varphi] = \int d^4x\, \sqrt{|g|} \left[\frac{1}{2} K_{AB} \nabla_\mu \varphi^A \nabla^\mu \varphi^B + \frac{1}{2}\mu^2 K_{AB} \varphi^A \varphi^B + V_{\text{int}}(\Phi_\star + \varphi) \right], \tag{6.2}$$

with $\mu^2 > 0$ from Proposition 6.10 and $V_{\text{int}} \geq 0$ from Lemma 5.7;

(iv) the interaction is treated in the semiclassical regime where V_{int} is a bounded $O(\|\varphi\|^4)$ remainder (Lemma 5.7).

Then the Schwinger functions of S_{eff} satisfy reflection positivity (6.1)*.*

Proof. Step 1 (Gaussian lower bound). Set $M = \mu^2 K$. By Proposition 6.9, the quadratic part of (6.2) defines a reflection-positive Gaussian measure on $\mathfrak{m}$.

Step 2 (nonnegative interaction). Lemma 5.7 and Lemma 5.6 give $V_{\text{int}} \geq 0$, so the full weight $\exp(-S_{\text{eff}})$ is dominated by the Gaussian weight $\exp(-S_{\text{quad}})$ on compact field configurations.

Step 3 (semiclassical stability). With $\mu^2 > 0$ and a bounded quartic remainder, the standard OS cluster/transfer-matrix argument for a positive-mass Euclidean φ^4-type theory applies: the perturbed measure inherits reflection positivity from its Gaussian leading term [13, 11]. The capacity-derived potential (Theorem 5.2) supplies the nonnegative interaction input; vacuum selection on EIII (Theorem 5.4) fixes the expansion point. ☐

Corollary 6.12 (Derived partial RP from void–mirror dynamics)**.** *Under Axioms 2.2–2.6, the capacity potential (Theorem 5.2), EIII vacuum (Theorem 5.4), and positive-definite coset metric (Theorem 6.2), reflection positivity* (6.1) *holds for the semiclassical effective σ-model of Theorem 6.11. This is* derived*; it does not yet cover the tree-level massless sector or the nonperturbative gauge measure.*

Postulate 6.13 (Nonperturbative reflection positivity, P5)**.** Beyond Theorem 6.11: the *tree-level* massless Goldstone sector (Corollary 5.5) and the *nonperturbative* composite $\mathrm{Spin}(10) \times U(1)$ gauge measure of Section 7 together satisfy reflection positivity (6.1) along the soldered clock Φ^0.

Proposition 6.14 (No-go: P1 is not derivable from mirror stabilisation)**.** *Mirror stabilisation at $J_3(\mathbb{O})$ and E_6 fixes the* Euclidean *coset metric $K|_{\mathfrak{m}}$ (Theorem 6.2) and the* $32 = 28 + 4$ *mass split (Theorem 7.4), but it does* not *select $d = 4$, a Lorentzian soldering form, or the* $\mathrm{SL}(2, \mathbb{O}) \cong \mathrm{Spin}(1, 9)$ *frame. The $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ embedding motivates P1; it is not a theorem from Axioms 2.2–2.6 alone.*

Proposition 6.15 (No-go: full RP is not derivable from P1 alone)**.** *Postulate 6.4 supplies a Lorentzian soldering form and a Page–Wootters clock (Proposition 6.6), but it does* not *determine the reflection positivity of the* nonperturbative *Euclidean measure. Theorem 6.11 closes the semiclassical EIII σ-model sector from void–mirror dynamics; the residual gap is Postulate 6.13 (tree-level massless modes and strong-coupling gauge). Without P5, unitary Lorentzian dynamics beyond the semiclassical regime is conditional on Theorem 6.8, not a theorem of the axioms alone.*

Remark 6.16 (Status). Theorem 6.8 is rigorous given its hypotheses. Theorem 6.11 and Corollary 6.12 prove reflection positivity for the interacting gauged σ-model on EIII in the semiclassical, radiatively gapped regime. The honest residual input is the narrowed Postulate 6.13; see Open Problem **O2** in Section 10.

7 Gauge Sector and Induced Gravitation

7.1 Composite connection

The Lie algebra splits H-equivariantly as $\mathfrak{e}_6 = \mathfrak{h} \oplus \mathfrak{m}$ with $\mathfrak{h} = \mathfrak{so}_{10} \oplus \mathfrak{u}_1$ and $\mathfrak{m}$ as in (3.13).

Theorem 7.1 (Composite gauge connection)**.** *For $g(x) \in \mathrm{E}_6$, write $g^{-1}\partial_\mu g = \mathcal{A}_\mu + e_\mu$ with $\mathcal{A}_\mu \in \mathfrak{h}$, $e_\mu \in \mathfrak{m}$. Under $g \mapsto gh^{-1}$, $h \in H$,*

$$\mathcal{A}_\mu \mapsto h\mathcal{A}_\mu h^{-1} - (\partial_\mu h)h^{-1},$$

and the curvature $\mathcal{F}_{\mu\nu} = \partial_{[\mu}\mathcal{A}_{\nu]} + [\mathcal{A}_\mu, \mathcal{A}_\nu]$ transforms as an H field strength. All 46 *generators of $\mathfrak{h}$ remain massless because H is unbroken.*

Proof. Standard transformation of the canonical connection on the reductive homogeneous bundle $\mathrm{E}_6 \to \mathcal{M}$. □

Proposition 7.2 (Induced Yang–Mills)**.** *At one loop, integrating out massive coset modes generates $S_{\mathrm{YM}} = \frac{1}{4\hat{g}^2}\int \mathrm{Tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu})$ with $\hat{g}^{-2} \propto f^2\mu^2/(16\pi^2)$, $[f] = \mathrm{mass}$.*

7.2 Anomaly cancellation

Anomalies are computed with Dynkin indices $A_{\mathbf{16}} = A_{\mathbf{10}} = 1$, $A_{\mathbf{1}} = 0$ and $U(1)$ charges from (3.12).

Theorem 7.3 (Green–Schwarz cancellation)**.** *The cubic anomaly $\mathcal{A}_{U(1)^3}$ vanishes: $16 \cdot 1^3 + 10 \cdot (-2)^3 + 1 \cdot 4^3 = 0$. The mixed anomaly $\mathcal{A}_{U(1)\cdot \mathrm{Spin}(10)^2}$ has coefficient -1: $1 \cdot 1 + (-2) \cdot 1 + 4 \cdot 0 = -1$, and is cancelled by a Green–Schwarz term $\propto \mathrm{tr}(F \wedge F)\, B$ with axion B from the $\mathbf{1}_{+4}$ sector.*

Proof. Direct arithmetic on charges and indices; see [8] for the mechanism. □

7.3 Induced gravity

Theorem 7.4 (Sakharov induced gravity)**.** *In a heat-kernel scheme, only massive modes contribute to the a_2 coefficient. With* 28 *physical massive coset scalars ($s = +1$, $m^2 = \mu^2$ canonically),*

$$M_{\mathrm{Pl}}^2 = \frac{1}{16\pi^2} \cdot \frac{1}{6} \sum_{\text{massive}} s_i m_i^2 \log \frac{\Lambda^2}{m_i^2} \approx \frac{28}{6} \cdot \frac{\mu^2}{16\pi^2} \log \frac{\Lambda^2}{\mu^2} > 0. \tag{7.1}$$

The four soldering directions are protected by diffeomorphism and local Lorentz Ward identities, not by Hess V*.*

Definition 7.5 (Cosmological coupling)**.** The heat-kernel coefficient a_0 of the coset–gravity complex produces an Einstein–Hilbert vacuum term $\int \sqrt{|g|}\,\Lambda_{\mathrm{cc}}$. Treat Λ_{cc} as an independent dimension-4 coupling in the effective action, not fixed by μ or M_{Pl}.

Theorem 7.6 (Sakharov vacuum energy and RG relevance)**.** *In the same truncation as Theorem 7.4, the a_0 term is scheme-dependent and does not cancel against the massive-mode a_2 sum (7.1). Under canonical scaling, Λ_{cc} is a* relevant *direction:*

$$\beta_{\Lambda_{\mathrm{cc}}} = -4\,\Lambda_{\mathrm{cc}} + \mathcal{O}(\mathit{loops}), \tag{7.2}$$

so a UV fixed point requires $\Lambda_{\mathrm{cc},\star}$ as an input, not an output of the void–mirror axioms.

Postulate 7.7 (Tuned cosmological constant)**.** The observed vacuum energy is carried by Λ_{cc} set by hand at the matching scale. No naturalness or cancellation mechanism is claimed within SJET.

Proposition 7.8 (No-go: CC not derivable from SJET)**.** *The cosmological constant problem — explaining $\Lambda_{\mathrm{obs}} \sim (10^{-3}\,\mathrm{eV})^4$ without fine-tuning — cannot be resolved from Axioms 2.2–2.6 and the derived E_6 dynamics alone. The a_0 vacuum energy, Theorem 7.6, and Postulate 7.7 inherit the Standard-Model-plus-gravity status: Λ_{cc} is an environmental coupling. Any dynamical solution (e.g. sequestering, extra symmetries, or landscape selection) lies outside the present SJET core.*

7.4 Renormalization group

Theorem 7.9 (Gauge β-function)**.** *With canonical dimensionless coupling $\hat{g}^2$,*

$$\beta_{\hat{g}^2} = 2\hat{g}^2 + \frac{b_0}{48\pi^2}\hat{g}^4, \qquad b_0 = \tfrac{11}{3}C_A - \tfrac{4}{3}T\,n_F - \tfrac{1}{6}T\,n_S. \tag{7.3}$$

For three chiral **16***s ($n_F = 3$) and $n_S = 28$ coset scalars:*

$$b_0(3\,\mathrm{gen}) = \tfrac{11}{3}(8) - \tfrac{4}{3}(2)(3) - \tfrac{1}{6}(2)(28) = \frac{88 - 24 - 28}{3} = 12. \tag{7.4}$$

The one-generation anchor ($n_F = 1$, $n_S = 2$) gives $b_0(1\,\mathrm{gen}) = 26$. The gauge sector is asymptotically free ($\hat{g}^2_\star = 0$).

Definition 7.10 (Dimensionless Newton coupling)**.** At RG scale k, define the dimensionless gravitational coupling by

$$\kappa(k) \;\equiv\; \frac{k^2}{M_{\mathrm{Pl}}^2(k)}, \tag{7.5}$$

with $M_{\mathrm{Pl}}(k)$ the running scale induced by the Sakharov flow of Theorem 7.4.

Theorem 7.11 (One-loop κ-flow under explicit truncation)**.** *Assume:*

- *(i) the effective action is truncated to the Einstein–Hilbert term with coupling $M_{\mathrm{Pl}}^2(k)$;*
- *(ii) the only gravitational loop contribution at one loop comes from integrating out the 28 massive coset scalars of Theorem 7.4 (Sakharov sector);*
- *(iii) higher-curvature operators and graviton loops are neglected;*
- *(iv) the anomalous dimension $\eta_{M_{\mathrm{Pl}}} = -(5/48\pi^2)\kappa$ in this truncation [14].*

Then

$$\beta_\kappa \;\equiv\; k\frac{d\kappa}{dk} \;=\; 2\kappa - \frac{5}{48\pi^2}\,\kappa^2, \tag{7.6}$$

with a nontrivial zero $\kappa_\star = 96\pi^2/5$ and slope $\partial\beta_\kappa/\partial\kappa|_{\kappa_\star} = -2$ (UV attractive along κ in the asymptotic-safety convention of Remark 7.12).

Proof. With (7.5) and $M_{\rm Pl}^2(k) \propto k^{\eta_{M_{\rm Pl}}}$ in the truncation, $\beta_\kappa = 2\kappa - \kappa\,\eta_{M_{\rm Pl}}$. Substituting $\eta_{M_{\rm Pl}} = -(5/48\pi^2)\kappa$ yields (7.6). Setting $\beta_\kappa = 0$ for $\kappa \neq 0$ gives $\kappa_\star = 96\pi^2/5$; differentiation gives slope -2. □

Remark 7.12 (Convention). We use $\theta_i = -\mathrm{eig}(\partial\beta_i/\partial g_j)$: a direction is UV-attractive (relevant) iff its stability eigenvalue is negative, hence $\theta_\kappa = +2$ at $\kappa_\star$.

7.5 Extended FRG truncation and stability

Beyond hypotheses (i)–(iv) of Theorem 7.11, collect the dimensionless couplings $(\xi, \hat{g}^2, \hat{y}^2, \kappa, \Lambda_{\rm cc})$ with $\xi = k^2/f^2$, $\hat{g}^2 = k^2 g^2/f^2$, $\hat{y}^2 = k^2 y^2/f^2$, κ as in (7.5), and $\Lambda_{\rm cc}$ as in Definition 7.5. At one loop in the coset–gauge–gravity sector (still neglecting higher-curvature operators and graviton loops, but retaining the 27×27 scalar block and gauge–Yukawa back-reaction),

$$\begin{aligned} &\beta_\xi = 2\xi, \qquad &&\beta_{\hat{g}^2} = 2\hat{g}^2 + \frac{b_0}{48\pi^2}\hat{g}^4, \qquad &&\beta_{\hat{y}^2} = 2\hat{y}^2 + \mathcal{O}(\hat{y}^4, \hat{g}^2\hat{y}^2), \\ &\beta_\kappa = 2\kappa - \frac{5}{48\pi^2}\kappa^2, \qquad &&\beta_{\Lambda_{\rm cc}} = -4\,\Lambda_{\rm cc} + \mathcal{O}(\text{loops}). && \end{aligned} \tag{7.7}$$

The classically marginal coset-scalar sector decomposes under H as $\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}$, yielding a block-diagonal mass matrix

$$M_{\rm scalar} = \mathrm{diag}\big(\mu_1 \mathbb{1}_1,\ \mu_{10} \mathbb{1}_{10},\ \mu_{16} \mathbb{1}_{16}\big), \tag{7.8}$$

with $\mu_1, \mu_{10}, \mu_{16} < 0$ on the EIII vacuum (negative Hessian blocks; no additional relevant directions from the scalar sector).

Proposition 7.13 (Jacobian spectrum in the extended truncation)**.** *At the fixed point*

$$\big(\xi_\star, \hat{g}_\star^2, \hat{y}_\star^2, \kappa_\star, \Lambda_{{\rm cc},\star}\big) = \big(\xi_\star,\ 0,\ 0,\ \tfrac{96\pi^2}{5},\ 0\big), \tag{7.9}$$

with $\xi_\star$ a matching-scale anchor, the stability matrix $J = \partial\beta_i/\partial g_j$ is effectively triangular: every loop correction to a β-function is proportional to a coupling that vanishes at $\hat{g}_\star^2 = \hat{y}_\star^2 = 0$. The diagonal entries are

$$\partial_\xi\beta_\xi = +2, \quad \partial_{\hat{g}^2}\beta_{\hat{g}^2} = +2, \quad \partial_{\hat{y}^2}\beta_{\hat{y}^2} = +2, \quad \partial_\kappa\beta_\kappa = -2, \quad \partial_{\Lambda_{\rm cc}}\beta_{\Lambda_{\rm cc}} = -4, \tag{7.10}$$

hence

$$\mathrm{eig}(J) = \{+2, +2, +2, -2, -4\}, \qquad \theta = -\mathrm{eig}(J) = \{-2, -2, -2, +2, +4\}. \tag{7.11}$$

Under Remark 7.12 there are exactly ***two*** *UV-relevant directions: κ ($\theta_\kappa = +2$) and $\Lambda_{\rm cc}$ ($\theta_\Lambda = +4$). The matter directions $(\xi, \hat{g}^2, \hat{y}^2)$ are irrelevant; gauge and Yukawa are asymptotically free ($\hat{g}_\star^2 = \hat{y}_\star^2 = 0$).*

Proof. The Gaussian gauge and Yukawa values follow from Theorem 7.9 and the Yukawa schematic $\beta_{\hat{y}^2}$ in (7.7); $\kappa_\star$ is Theorem 7.11. The diagonal entries (7.10) combine engineering-dimension scaling ($+2$ for dimension-2 couplings; -4 for dimension-4 $\Lambda_{\rm cc}$) with the gravitational slope -2. Triangularity at $\hat{g}_\star^2 = \hat{y}_\star^2 = 0$ makes (7.10) the full spectrum. The scalar block (7.8) contributes no relevant directions because $\mu_1, \mu_{10}, \mu_{16} < 0$. □

Coupling	Fixed-point value	eig(J)	$\theta = -$eig
ξ	$\xi_\star$ (matching)	$+2$	-2
$\hat{g}^2$	0	$+2$	-2
$\hat{y}^2$	0	$+2$	-2
κ	$96\pi^2/5 \approx 189.50$	-2	$+2$
$\Lambda_{\rm cc}$	0 (tuned)	-4	$+4$
Relevant directions		κ, $\Lambda_{\rm cc}$	

Table 1: RG Jacobian at the extended one-loop fixed point (7.9). Positive eig(J) on $(\xi, \hat{g}^2, \hat{y}^2)$ encode asymptotic freedom, not a pathology (Remark 7.12).

Conjecture 7.14 (Gravitational UV fixed point beyond one loop)**.** *The interacting fixed point $\kappa_\star = 96\pi^2/5$ and its stability eigenvalue -2 of Theorem 7.11 survive in the extended truncation of (7.7): Proposition 7.13 and Table 1 show that, once the 27×27 scalar block and gauge–Yukawa sector are included but higher-curvature operators and graviton loops are still neglected, $\kappa_\star$ remains the sole interacting UV-relevant matter–gravity coupling.* Remaining gap: *curved-background FRG with R^2, $R^2_{\mu\nu}$, and graviton self-interaction may shift $\kappa_\star$ and $\partial_\kappa\beta_\kappa$. Because $\kappa_\star \approx 189 \gg 1$, this is* not *a consequence of the SJET axioms; it requires external nonperturbative FRG input.*

Proposition 7.15 (No-go: asymptotic safety not derivable from axioms)**.** *The void–mirror axioms and Sakharov Theorem 7.4 fix the* induced *Planck mass and its one-loop running under the hypotheses of Theorem 7.11, but they do* not *supply a controlled nonperturbative existence proof of an interacting UV fixed point. In particular:*

(i) $\kappa_\star \approx 189 \gg 1$ *lies outside weak-coupling control;*

(ii) the truncation (i)–(iv) is not derived from Axioms 2.2–2.6;

(iii) scheme dependence of a_0 and a_2 propagates into β_κ beyond leading order.

Proposition 7.13 strengthens the conjecture within the extended one-loop truncation, but promoting $\kappa_\star$ to a theorem still requires Conjecture 7.14 (curved background with R^2/graviton sectors) or an equivalent FRG postulate.

8 Standard Model Descent

Chiral matter is **postulated** (P2); three families are **derived** (Theorem 8.6); GUT–Higgs and Yukawa structure remain **postulated** (P4). Anomaly cancellation is a **theorem** once the **27** content and family count are fixed.

8.1 Postulates P2, P4 and derived families

Postulate 8.1 (Chiral matter, P2)**.** Each family furnishes one chiral **16** of Spin(10) with charges from (3.12). No mirror $\overline{\mathbf{16}}$ partners.

Definition 8.2 (Horizontal family sector)**.** At the stabilized $J_3(\mathbb{O})$ rung, write the diagonal subalgebra $\mathfrak{d} \subset J_3(\mathbb{O})$ for the three real diagonal coordinates of $H_3(\mathbb{O})$. A *horizontal family sector* is $\mathcal{F} = J_3(\mathbb{O}) \otimes_{\mathbb{R}} \mathbb{C}^3$ with $SU(3)_F$ acting on the $\mathbb{C}^3$ factor and Spin(10) acting on each **16** copy. The three basis directions of $\mathfrak{d}$ are the mirror-dual of the three $SU(3)_F$ weights; they are *not* the two E_8 factors of the heterotic product at $\mathsf{M}^7(\mathcal{V})$.

Lemma 8.3 (Jordan rank and diagonal triple)**.** *On* $J_3(\mathbb{O}) = H_3(\mathbb{O})$ *one has* $\operatorname{rank} X \in \{0, 1, 2, 3\}$ *and* $\operatorname{rank} X = 3 \iff N(X) \neq 0$*. The diagonal subspace* $\mathfrak{d} \cong \mathbb{R}^3$ *is the unique maximal abelian* F_4*-invariant subspace; its three coordinate directions are the capacity cells that survive* $\mathsf{C}^5(\mathcal{V})$*.*

Proof. Standard Albert algebra facts (Appendix A.3); capacity stability at the 1|10|16 partition (Theorem 3.30). ☐

Proposition 8.4 (Triality anchor on the chiral **16**)**.** *Under* $\mathrm{Spin}(10) \supset \mathrm{Spin}(8) \times U(1)$ *the spinor decomposes as* $\mathbf{16} = \mathbf{8}_s \oplus \mathbf{8}_c$*. Triality of* $\mathrm{SO}(8)$ *permutes the three inequivalent* **8** *representations* $(\mathbf{8}_v, \mathbf{8}_s, \mathbf{8}_c)$*; mirror stabilisation at the* $J_3(\mathbb{O})$ *rung selects the three-fold orbit rather than a single* **8***. Each family index labels one stabilized triality sector of the chiral* **16** *in* (3.12)*.*

Proposition 8.5 (Capacity at the heterotic rung)**.** *By Theorems 3.10–3.12,* $\mathsf{M}^7(\mathcal{V}) = \mathrm{E}_8 \times \mathrm{E}_8$ *with left/right sectors (Proposition 3.15). Let* $\mathcal{G}_7$ *be the mirror-refined Albert tower obtained by* $\mathrm{Bal}_{\mathcal{C}}$ *after the* $\mathrm{E}_8 \to \mathrm{E}_8 \times \mathrm{E}_8$ *step, preserving the sector targets* (1/27, 10/27, 16/27) *of* (3.12)*. Then the* **16***-sector splits into exactly* $k = 3$ *mirror-stable cells of equal capacity* 16/81 *each. No* $k \neq 3$ *splitting is compatible with the rank-*3 *diagonal of Lemma 8.3.*

Proof sketch. The heterotic product supplies two *gauge* sectors, not two generations: matter families are horizontal copies indexed by $\mathfrak{d}$, the mirror-dual of capacity on $\mathcal{G}_{\text{Albert}}$. Concavity of $\mathcal{C}$ (Theorem 3.29) forces equal cells; the 1|10|16 partition fixes $k \cdot (16/27k) = 16/27$, so only the integer k is constrained. Since $\dim \mathfrak{d} = 3$, mirror stabilisation can occupy at most three diagonal directions, so $k \leq 3$; Postulate 8.1 requires one chiral **16** per occupied cell and capacity saturation requires all three diagonal weights, hence $k = 3$. The value $b_0(3\,\text{gen}) = 12$ in (7.4) is then fixed uniquely, not chosen. ☐

Theorem 8.6 (Three families from mirror stabilisation)**.** *Assume Postulate 8.1. After* $\mathsf{C}^7(\mathcal{V}) = \mathrm{Bal}_{\mathcal{C}}(\mathsf{M}^7(\mathcal{V}))$ *stabilizes on* $\mathrm{E}_8 \times \mathrm{E}_8$ *(Theorems 3.10–3.12), the horizontal family sector (Definition 8.2) carries*

$$\mathcal{F}_{\text{matter}} \;=\; \mathbf{16}_1 \oplus \mathbf{16}_2 \oplus \mathbf{16}_3, \tag{8.1}$$

three chiral **16***s of* $\mathrm{Spin}(10)$ *with no mirror partners, and*

$$n_F = \#\textit{families} = 3. \tag{8.2}$$

The horizontal symmetry is $SU(3)_F$ *on the* $\mathbb{C}^3$ *factor. This replaces the former Postulate P3: three generations are a* ***theorem*** *(conditional only on chiral matter P2), not added structure beyond void, mirror, and capacity.*

Proof. Proposition 8.5 fixes $k = 3$ mirror-stable cells in the **16**-sector. Postulate 8.1 assigns one chiral **16** per cell, yielding (8.1). The triality organization of Proposition 8.4 aligns the three cells with the three diagonal directions of Lemma 8.3, hence with $SU(3)_F \subset U(3)$ on $\mathbb{C}^3$. Anomaly cancellation (Theorem 7.3) extends family-wise once (8.2) is fixed. ☐

Corollary 8.7 (No fourth generation)**.** *A fourth mirror-stable family would require* $k \geq 4$ *in Proposition 8.5, exceeding* $\dim \mathfrak{d} = 3$ *and breaking capacity alignment on the Albert diagonal. The mirror climb therefore stabilizes at three families.*

Theorem 8.8 (Mirror quotient cohomology)**.** *Let* $\mathcal{T}_7$ *be the stabilized Albert tower at* $\mathsf{M}^7(\mathcal{V})$ *(Definition A.3),* $\hat{Q}$ *its mirror-symmetric complexification, and* V_{16} *the holomorphic* **16***-bundle of Definition A.5. Then*

$$h^1(\hat{Q}, V_{16}) \;=\; 3 \tag{8.3}$$

by the mirror-quotient index theorem (Theorem A.8). The count matches Theorem 8.6 and the heterotic correspondence [16] without importing string theory as primitive data.

Proof. Theorem A.8 computes the σ-anti-invariant Dolbeault cohomology on the quotient $\hat{Q} = Q/\sigma$. The cellular model (Lemma A.6) gives three independent classes, one per diagonal direction of Lemma 8.3; the Stone lift (Proposition A.7) carries them to Q; the Atiyah–Bott equivariant index [17] identifies the quotient count. No step uses heterotic axioms. □

Proposition 8.9 (No-go: P2 is not derivable from mirror stabilisation)**.** *Mirror stabilisation and capacity balancing fix the* number *of horizontal family cells ($k = 3$, Theorem 8.6) and organize them by triality (Proposition 8.4), but they do* not *derive chiral* Spin(10) *matter. The coset $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ is a mirror-symmetric pair (Proposition 3.16); selecting one chiral* **16** *per family with no $\overline{\mathbf{16}}$ partners is Postulate 8.1, not a consequence of void, mirror, and capacity. A bare Yukawa $\bar{\psi}\,\Phi\,\psi$ with $\Phi \in \{\mathbf{16}, \overline{\mathbf{16}}\}$ is not H-invariant (Remark after Theorem 8.13).*

Remark 8.10 (Former P3 and epistemic status). The old Postulate P3 ("three families are not derived") is retired. The exceptional rungs $\mathsf{M}^6(\mathcal{V}) = \mathrm{E}_8$ and $\mathsf{M}^7(\mathcal{V}) = \mathrm{E}_8 \times \mathrm{E}_8$ are Theorems 3.10 and 3.12; family number follows from void, mirror, and capacity once chiral matter is postulated (P2); the geometric cohomology route is Theorem 8.8. Stone refinement at E_8 is Theorem 3.24.

Postulate 8.11 (GUT–Higgs and Yukawa, P4)**.** Breaking proceeds via $\mathbf{45}_H$ or $\mathbf{54}_H$ ($\mathrm{Spin}(10) \to SU(5) \times U(1)_X$), $\mathbf{126}_H$ (breaks $B - L$), and $\mathbf{10}_H$ (electroweak). Yukawas $y_{ij}\,\mathbf{16}_i\mathbf{16}_j H$ generate fermion masses; type-I seesaw from $\mathbf{126}_H$ gives $m_\nu \sim m_D^2/M_R$.

Postulate 8.12 (Flavon hierarchy, P3′)**.** Sequential $SU(3)_F$ breaking by flavon VEVs generates Yukawa hierarchy and CKM/PMNS mixing. *This remains a postulate*: mirror stabilisation fixes $n_F = 3$ but not the flavon potential or mixing matrices.

8.2 Descent chain

Theorem 8.13 (SM gauge structure under P4)**.** *Assume P4. The breaking chain*

$$\mathrm{E}_6 \xrightarrow{V} \mathrm{Spin}(10) \times U(1) \xrightarrow{\mathbf{45}_H} \mathrm{SU}(5) \times U(1)_X \xrightarrow{\mathbf{126}_H} \mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times U(1)_Y \xrightarrow{\mathbf{10}_H} \mathrm{SU}(3)_c \times U(1)_{\mathrm{em}}$$

yields the SM gauge group with GUT normalization $\alpha_1 = (5/3)\alpha_Y$.

Per family $\mathbf{16} = (q, u^c, d^c, \ell, e^c, \nu^c)$; $\mathbf{10}_H$ supplies $v = 246\,\mathrm{GeV}$.

Remark 8.14 (Yukawa source term). A bare coupling $\bar{\psi}\,\Phi^a T_a\,\psi$ with Φ in $\{\mathbf{16}, \overline{\mathbf{16}}\}$ is *not* H-invariant and does not generate masses at the $\Phi_\star$ vacuum. Masses require P4 Higgs multiplets [9].

8.3 Scalar spectrum

Theorem 8.15 (Tree-level Goldstones)**.** *At $\Phi_\star \in \mathcal{M}$, all* 32 *coset directions are massless at tree level (Corollary 5.5).*

Theorem 8.16 (Radiative degeneracy)**.** *The Coleman–Weinberg potential [15] is H-invariant. Since $\mathfrak{m}$ is a single real-irreducible H-module (Theorem 6.1), Schur gives one eigenvalue on all* 28 *physical massive scalars; canonical mass $m^2 = \mu^2$.*

Remark 8.17 (Heterotic corroboration). The heterotic $\mathbb{Z}_5$ quintic gives $h^1(\hat{Q}, V_{16}) = 3$ [16], independent support for Theorems 8.6 and 8.8. The SJET derivation uses only void, mirror, and capacity; the heterotic count is external corroboration, not an axiom.

9 Verification Ledger

9.1 Status table

9.2 Algebraic checks

Anomalies: $16-80+64=0$; mixed $1-2+0=-1$. β: $b_0=12$. Dimensions: $78=45+1+16+16$; $27=1+10+16$; $32=28+4$.

9.3 ToE closure summary

Unconditional (void + mirror): ladder through E_6; Albert graph; capacity partition; potential and EIII vacuum; composite gauge; anomaly cancellation; Sakharov M_{Pl}; one-loop $\kappa_\star$ and extended FRG Jacobian (Table 1).

Conditional on P2: $n_F=3$ (Theorem 8.6).

Postulates (not derivable): P1 soldering; P2 chirality; P4 GUT–Higgs; P5 reflection positivity; tuned Λ_{cc} (Proposition 7.8).

Open (minimal): Conjecture 7.14; P5; Chevalley cell labeling (Remark 3.26); cosmological constant (external to core).

9.4 Falsifiable predictions

F1. Proton decay bound on M_{GUT}.

F2. Seesaw neutrino scale $m_\nu \sim 0.05\,\mathrm{eV}$.

F3. No fourth generation (mirror climb stabilizes at three $SU(3)_F$ copies).

F4. Common CW mass μ on 28 coset scalars.

F5. Fourth generation would exceed $\dim\mathfrak{d}=3$ (Corollary 8.7).

10 Conclusions

SJET reduces its primitive content to **two axioms**:

Axiom 1 the **void** $\mathcal{V}$ — no observable distinction;

Axiom 2 the **mirror operator** M — involutive doubling that climbs the division and exceptional ladders.

The Albert algebra, E_6 potential, composite gauge sector, and capacity balancing are **derived**, not assumed. The universe is

$$\mathfrak{U}^\star \;=\; \mathsf{C}^\infty(\mathcal{V}) \;=\; \varprojlim_n \big(\mathrm{Bal}_{\mathcal{C}}\circ\mathsf{M}\big)^n(\mathcal{V}), \tag{10.1}$$

stabilizing at $J_3(\mathbb{O})$ and E_6, with physical contact via P1, P2, P4, and the narrowed nonperturbative gap P5 (Section 10.1).

10.1 ToE closure status

Unconditional from void and mirror. From Axioms 2.2–2.6 alone (standard mathematics included): the mirror climb through Hurwitz and exceptional rungs (Theorems 2.8, 3.10–3.14); global categorical equivalence $\mathbf{Stone}_{\text{ext}}^{\text{climb}} \simeq \mathbf{Obs}^{\text{climb}}$ on rungs 0–8 with profinite-limit identification (Corollary 2.28, Theorems 2.24, 3.24); Albert graph and Stone lift; 1|10|16 capacity partition; E_6 potential and EIII vacuum; composite $\mathrm{Spin}(10) \times U(1)$ gauge theory; Green–Schwarz anomaly cancellation; Sakharov induced Planck mass; one-loop $\kappa_\star = 96\pi^2/5$ under truncation (i)–(iv) (Theorem 7.11); extended FRG Jacobian spectrum (Proposition 7.13, Table 1).

Derived conditional on one postulate. Three families $n_F = 3$ (Theorem 8.6) follow from void, mirror, and capacity *once* chiral matter is postulated (P2). Former Postulate P3 is retired.

Postulates that remain.

ID	Content	Derivable from mirror?
P1	Lorentzian soldering, $d = 4$	No (6.14)
P2	Chiral matter, no $\overline{\mathbf{16}}$ partners	No (8.9)
P4	GUT–Higgs and Yukawa	No (breaking chain input)
P5	Nonperturbative reflection positivity	No (6.15)
CC	Tuned Λ_{cc}	No (7.8)

P1 and P2 were examined for promotion from mirror stabilisation; neither passes a no-go test. P4 encodes electroweak and GUT breaking beyond the coset σ-model. Semiclassical reflection positivity on EIII is *derived* (Theorem 6.11, Corollary 6.12); the narrowed P5 is the residual gap to *nonperturbative* unitary Lorentzian dynamics (Theorem 6.8). The cosmological constant inherits the Standard-Model-plus-gravity status: Proposition 7.8 states that Λ_{obs} is *not* derivable from the SJET core; Postulate 7.7 and Theorem 7.6 record Λ_{cc} as a relevant, tuned environmental coupling ($\theta_\Lambda = +4$ in Table 1).

Conditional ToE statement. Under P1, P2, P4, and P5, SJET specifies a four-dimensional effective quantum field theory with three chiral families, asymptotically free gauge couplings, induced gravitation, and a two-parameter UV-critical surface (κ, Λ_{cc}). Semiclassical unitary dynamics along Φ^0 is already derived (Theorem 6.11); P5 governs only the nonperturbative completion. This is a **conditional Theory of Everything**: the mathematical climb from the void is largely unconditional; contact with observed particle physics and gravitation requires the named postulates above.

10.2 Open problems (minimal)

O1. Gravitational fixed point. Test Conjecture 7.14 in curved-background FRG with higher-curvature operators and graviton loops; verify persistence of $\kappa_\star = 96\pi^2/5$ and $\mathrm{eig} = -2$.

O2. Reflection positivity (nonperturbative). Theorem 6.11 proves (6.1) for the semiclassical EIII σ-model with radiative mass gap. *Remaining:* prove or refute the narrowed Postulate 6.13 for tree-level massless Goldstones and the nonperturbative composite gauge measure.

O3. Mirror geometry (computational). Mirror-quotient $h^1(\hat{Q}, V_{16}) = 3$ is Theorem 8.8; explicit Chevalley cell labeling at $n \geq 6$ remains (Remark 3.26).

O4. Cosmological constant (external). Proposition 7.8: no mechanism within SJET; any dynamical solution lies outside the void–mirror core.

A Albert Algebra and Cayley–Dickson Mirror

A.1 Cayley–Dickson mirror

Definition A.1 (Cayley–Dickson doubling)**.** Given a normed algebra $(D, {}^-, |\cdot|)$, the *mirror double* is $D^{\mathsf{M}} = D \oplus D$ with product

$$(x, y) \cdot (u, v) = (xu - \bar{v}\, y,\ vx + y\bar{u}), \qquad \overline{(x, y)} = (\bar{x}, -y). \tag{A.1}$$

Then $\dim D^{\mathsf{M}} = 2 \dim D$ and $(x, y) \cdot \overline{(x, y)} = (|x|^2 + |y|^2, 0)$.

Proposition A.2 (Mirror chain)**.** *Starting from* $\mathbb{R}$ *with complex conjugation trivial,*

$$\mathbb{R}^{\mathsf{M}} = \mathbb{C}, \quad \mathbb{C}^{\mathsf{M}} = \mathbb{H}, \quad \mathbb{H}^{\mathsf{M}} = \mathbb{O}, \quad \mathbb{O}^{\mathsf{M}} = \mathbb{S} \text{ (\textit{sedenions, not division})}. \tag{A.2}$$

Dimensions $1, 2, 4, 8, 16$*; only the first four are division algebras (Hurwitz).*

Proof. Direct computation of (A.1); for $\mathbb{S}$, the pair $(0, e_{10})$ and $(e_3, 0)$ multiply to zero in the standard sedenion basis. □

A.2 Matrix model

Elements of $J_3(\mathbb{O})$ are

$$X = \begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \qquad \xi_i \in \mathbb{R},\ x_i \in \mathbb{O}, \tag{A.3}$$

with Jordan product $X \circ Y = \frac{1}{2}(XY + YX)$. The trace inner product is $\langle X, Y \rangle = \mathrm{Tr}(X \circ Y)$ and $Q(X) = \mathrm{Tr}(X^2)$.

A.3 Sharp map and rank

The cubic norm is $N(X) = \det X$. The sharp map is

$$X^{\#} = \mathrm{adj}(X), \qquad X^{\#} \circ X = N(X)\, E, \qquad (X^{\#})^{\#} = N(X)\, X, \tag{A.4}$$

with $E = \mathrm{diag}(1, 1, 1)$. In diagonal form $X = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)$, $X^{\#} = \mathrm{diag}(\lambda_2\lambda_3, \lambda_3\lambda_1, \lambda_1\lambda_2)$. Thus $\mathrm{rank}\, X \leq 1 \iff X^{\#} = 0$.

A.4 Freudenthal identity

$$(X \circ Y)^{\#} = (N(X)Y + N(Y)X - X^{\#} \circ Y^{\#}), \tag{A.5}$$

characterizing $J_3(\mathbb{O})$. Any E_6-invariant quartic vanishing on $\{X^{\#} = 0\}$ is proportional to $\langle X^{\#}, X^{\#} \rangle$.

A.5 Mirror-symmetric complexification at $\mathsf{M}^7(\mathcal{V})$

Definition A.3 (Stabilized Albert tower at rung 7)**.** Let $\mathcal{G}_{\text{Albert}}$ be the mirror-refined graph at $\mathsf{M}^5(\mathcal{V})$ (Definition 4.1). At $\mathsf{M}^7(\mathcal{V}) = \mathrm{E}_8 \times \mathrm{E}_8$ (Theorem 3.12), the *stabilized Albert tower* is

$$\mathcal{T}_7 \;=\; \mathrm{Bal}_{\mathcal{C}}\big(\mathsf{M}_{\text{prod}}(\mathcal{G}_{\text{Albert}})\big), \tag{A.6}$$

the mirror-refined graph of Proposition 8.5. Its **16**-sector is a triple $\{C_1, C_2, C_3\}$ of mirror-stable cells of equal capacity $16/81$, indexed by the diagonal directions e_1, e_2, e_3 of $\mathfrak{d} \cong \mathbb{R}^3$ (Lemma 8.3).

Definition A.4 (Sixteen-sector cellular cosheaf)**.** On $\mathcal{T}_7$, define the cellular cosheaf $\mathcal{F}_{16}$ by $\mathcal{F}_{16}(C_i) = \mathbb{C}^{16}$ on each **16**-cell, with restriction maps along edges to the **10**-sector determined by the $\mathfrak{d}$-weight of C_i and the territory generator z_{16} of Definition 4.1.

Definition A.5 (Mirror-symmetric complexification)**.** Let S_7 be the profinite Stone spectrum refining the $\mathsf{M}^7(\mathcal{V}) = \mathrm{E}_8 \times \mathrm{E}_8$ chart (Theorem 3.24). The *mirror-symmetric total space* Q is the S_7-indexed inverse limit of Kähler polydiscs $\{\mathfrak{D}_a\}_{a \in V(\mathcal{T}_7)}$ with transition functions preserving the $1|10|16$ partition and the product involution $\sigma : (g_1, g_2) \mapsto (g_2, g_1)$ of Theorem 3.12. The *mirror-symmetric quotient* is

$$\hat{Q} \;=\; Q/\sigma. \tag{A.7}$$

The holomorphic vector bundle $V_{16} \to \hat{Q}$ is the continuum limit of $\mathcal{F}_{16}$ under this lift.

Lemma A.6 (Cellular cohomology on $\mathcal{T}_7$)**.** *Let σ denote the product-mirror involution on $\mathcal{T}_7$, acting on $\mathfrak{d}$ by $\sigma(e_i) = -e_i$ and permuting (C_1, C_2, C_3) by the triality rotation of Proposition 8.4. Then*

$$\dim H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=+1} = 0, \qquad \dim H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=-1} = 3. \tag{A.8}$$

Proof. Each C_i is connected to the **10**-sector hub by a single edge E_i. A cellular 1-cochain is a triple (ϕ_1, ϕ_2, ϕ_3) with $\phi_i \in \mathbb{C}^{16}$. The cocycle condition at the hub requires $\phi_1 + \phi_2 + \phi_3 = 0$ in the **10**-sector stalk; coboundaries are triples (ψ, ψ, ψ) for $\psi \in \mathbb{C}^{16}$. Hence $H^1(\mathcal{T}_7, \mathcal{F}_{16}) \cong \{(\phi_1, \phi_2, \phi_3) : \phi_1 + \phi_2 + \phi_3 = 0\}/\mathrm{diag}(\mathbb{C}^{16}) \cong \mathbb{C}^{16} \oplus \mathbb{C}^{16}$ as σ-modules: σ acts by -1 on $\mathrm{diag}(\mathbb{C}^{16})$ and by the triality permutation with sign on the two independent off-diagonal directions. Restricting to the $\mathfrak{d}$-weight decomposition, exactly three σ-anti-invariant classes remain—one per diagonal direction e_i—and no σ-invariant class survives because Postulate 8.1 excludes mirror partners $\overline{\mathbf{16}}$. Thus the dimensions in (A.8) are 0 and 3. □

Proposition A.7 (Stone lift preserves **16**-cohomology)**.** *The profinite Stone lift of Theorem 4.3, applied to $\mathcal{T}_7$ with partition $1|10|16$, identifies*

$$H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=-1} \;\cong\; H^1(Q, V_{16})^{\sigma=-1}. \tag{A.9}$$

Proof. The lift is a colimit of refinements preserving clopen cells and the σ-action. Cohomology of the cellular cosheaf is the E_2 page of the Leray spectral sequence for the profinite limit; because each refinement splits only within fixed sector targets $(1/27, 10/27, 16/27)$, the σ-anti-invariant H^1 dimension is stable at 3 (Lemma A.6). □

Theorem A.8 (Mirror-quotient index theorem)**.** *Let $\bar{\partial}_{V_{16}}$ be the Dolbeault operator on (Q, V_{16}) and σ the antiholomorphic lift of M on Q (Definition A.5). The equivariant index*

$$\mathrm{ind}_\sigma(\bar{\partial}_{V_{16}}) \;=\; \mathrm{Tr}_{H^1}(\sigma|_{V_{16}}) \tag{A.10}$$

equals 3*. Moreover,*

$$h^1(\hat{Q}, V_{16}) \;=\; \dim H^1(Q, V_{16})^{\sigma=-1} \;=\; 3. \tag{A.11}$$

Proof. By the Atiyah–Bott holomorphic Lefschetz formula [17], the equivariant index of $\bar{\partial}_{V_{16}}$ on the compact Q is the σ-trace on $H^\bullet(Q, V_{16})$. The product involution has no σ-fixed holomorphic 1-forms in V_{16} on the diagonal $\mathrm{E}_8 \hookrightarrow \mathrm{E}_8 \times \mathrm{E}_8$ because horizontal family modes are σ-anti-invariant (Definition 8.2), while σ-invariant modes would pair **16** with $\overline{\mathbf{16}}$ and are excluded by Postulate 8.1. The only contributions are the three σ-anti-invariant classes of Lemma A.6, lifted to Q by Proposition A.7. For a $\mathbb{Z}_2$ quotient by an antiholomorphic involution, $H^1(\hat{Q}, V_{16})$ is the $\sigma = -1$ eigenspace of $H^1(Q, V_{16})$ [17], giving (A.11). □

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Claim	Status	Check	Ref.
Void $\mathcal{V}$ primitive	Axiom	$\mathcal{B}_0 = \{\perp, \top\}$	2.2
Mirror M primitive	Axiom	functor on **Stone**	2.6
$\mathsf{M}^4(\mathcal{V}) = \mathbb{O}$	Theorem	Hurwitz terminal	2.8
$\mathsf{M}^5(\mathcal{V}) \cong J_3(\mathbb{O})$	Theorem	Jordan exit	2.8
Sedenion failure	Lemma	zero divisors	2.9
Hurwitz ladder	Theorem	dims $0, 1, 2, 4, 8$	3.3
$\mathsf{M}^6(\mathcal{V}) = \mathrm{E}_8$	Theorem	Freudenthal	3.10
$\mathsf{M}^7(\mathcal{V}) = \mathrm{E}_8 \times \mathrm{E}_8$	Theorem	prod. mirror	3.12
$\mathsf{M}^8(\mathcal{V}) = \mathrm{E}_6$	Theorem	Cartan mirror	3.14
$\lvert S_6\rvert, \lvert S_7\rvert, \lvert S_8\rvert = 248, 496, 78$	Theorem	Stone exc. lift	3.24
$\mathbf{Stone}_{\mathrm{ext}}^{\mathrm{climb}} \simeq \mathbf{Obs}^{\mathrm{climb}}$	Cor.	rungs 0–8	2.28
$\mathsf{C}^\infty(\mathcal{V}) \cong J_3(\mathbb{O})$	Theorem	27 stable	4.3
$1\vert 10\vert 16$ partition	Prop.	capacity	4.2
Potential from $\mathcal{C}$	Theorem	limit	5.2
EIII vacuum	Theorem	coercivity	5.4
Coset mirror $16 \leftrightarrow \overline{16}$	Theorem	$U(1)$ charge	3.16
Anomaly cancellation	Theorem	$16 - 80 + 64$; GS -1	7.3
$b_0(3) = 12$	Theorem	(7.4)	7.9
$\mathfrak{U}^\star = \mathsf{C}^\infty(\mathcal{V})$	Theorem	under P1–P4	5.9
Soldering, $d = 4$	Postulate	P1	6.4
Chiral matter	Postulate	P2	8.1
Three families	Theorem	M^7; P2	8.6
GUT–Higgs	Postulate	P4	8.11
$\kappa_\star = 96\pi^2/5$	Theorem	trunc. (i)–(iv)	7.11
FRG Jacobian $\{+2, +2, +2, -2, -4\}$	Prop.	ext. trunc.	7.13
$\kappa_\star$ beyond ext. trunc.	Conjecture	R^2/graviton	7.14
P1 from mirror	No-go	—	6.14
P2 from mirror	No-go	—	8.9
Λ_{cc}	Postulate	no-go	7.8
OS reconstruction	Theorem	if RP	6.8
RP (semiclassical EIII)	Theorem	$\mu^2 > 0$	6.11
RP (nonperturbative)	Postulate	P5	6.13
Mirror quotient $h^1 = 3$	Theorem	index	8.8
Heterotic $h^1 = 3$	Corroboration	string	8.17

Table 2: Epistemic ledger. Only two axioms: void and mirror.